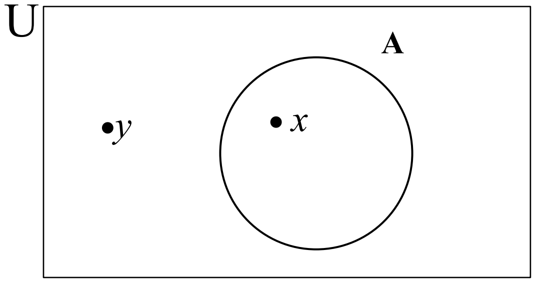
Chapter 1 Fundamentals

<u>Set</u>

- A set is a well-defined collection of objects.
- $x \in A$ means x is an element or belongs to A.



element $\in$ set

belongs to

$x \in A$

$y \notin A$

not belong to

Eg 1: Let A = {1, 3, 5, 7}.Then $1 \in A$, $3 \in A$, but $2 \notin A$.

Eg 2: The set consisting of all letters in the word “byte” can be expressed by {b, y, t, e}

Eg 3: a) $Z^+ = \{x \mid x \text{ is a positive integer}\}$
$= \{1, 2, 3, \ldots\}$

b) $N = \{x \mid x \text{ is a natural number}\}$
$= \{0, 1, 2, 3, \ldots\}$

c) $Z = \{x \mid x \text{ is an integer}\}$
$= \{\underbrace{\ldots, -3, -2, -1}_{Z^-}, 0, \underbrace{1, 2, 3, \ldots}_{Z^+}\}$

d) $Q = \{x \mid x \text{ is a rational number}\}$
$= \{\frac{a}{b}, b \neq 0\}$

e) R = $\{x \mid x$ is a real number$\}$

f) The set that has no elements in (called empty set), it is denoted by $\{\}$ or $\varnothing$.

Eg 4: Since the square of a real number is always non-negative,
$\{x \mid x$ is a real number and $x^2 = -1\}$ = $\{\ \}$ or ϕ
no solution

Eg 5: If A = $\{1, 2, 3\}$ and
B = $\{x \mid x$ is a positive integer and $x^2 < 12\}$,
then A __=__ B.

B = $\{1, 2, 3\}$

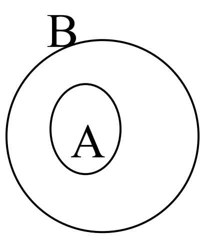
$x=1,\ 1^2=1<12$
$x=2,\ 2^2=4<12$
$x=3,\ 3^2=9<12$
$x=4,\ 4^2=16 \not< 12$

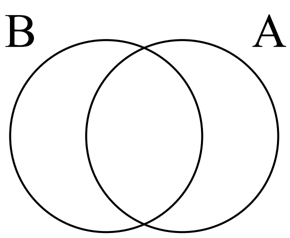
Eg 6: If A = {JAVA, PASCAL, C++} and
B = {C++, JAVA, PASCAL},
then A __=__ B.

set $\subseteq$ set

Subsets $\subseteq$

➢ If every element of A is also an element of B, then A is a subset of B.





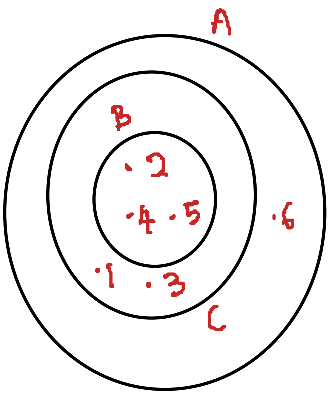
A __$\subseteq$__ B

A __$\nsubseteq$__ B

1,2,3...　　...,-1,0,1...

Eg 7: Z^+ __$\subseteq$__ Z　　　Z __$\subseteq$__ Q

Eg 8: Let A = {1, 2, 3, 4, 5, 6}, B = {2, 4, 5} and
C = {1, 2, 3, 4, 5}



B $\subseteq$ A B $\subseteq$ C C $\subseteq$ A

A_____B C_____B A_____C

Eg 9: If A is any set, then $A \subseteq A$.

- Every set is a subset to itself.

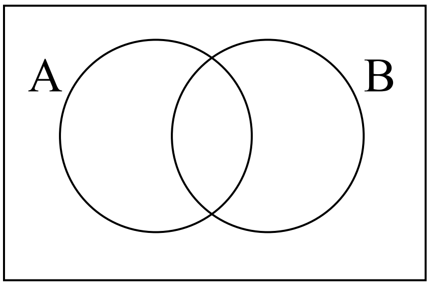
Eg 10: Let A = {1, 2, 3}

Subsets of A:

Power set of A:

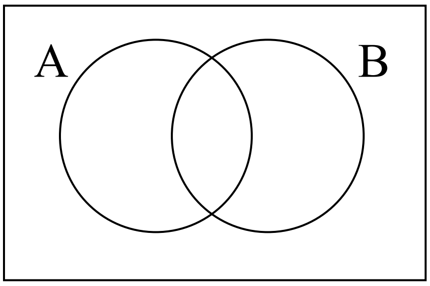
Operations on Sets

1) $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$



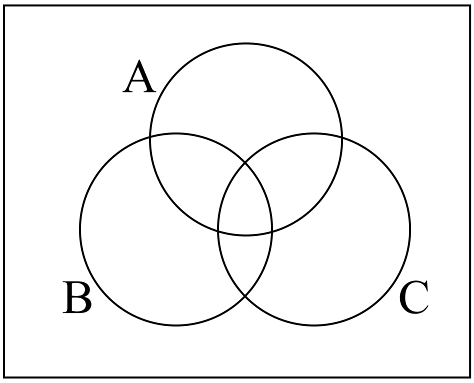
Eg 11: Let A = {a, b, c, e, f} and B = {b, d, r, s}. Find $A \cup B$.

2) $\mathbf{A \cap B} = \{x \mid x \in A \text{ and } x \in B\}$

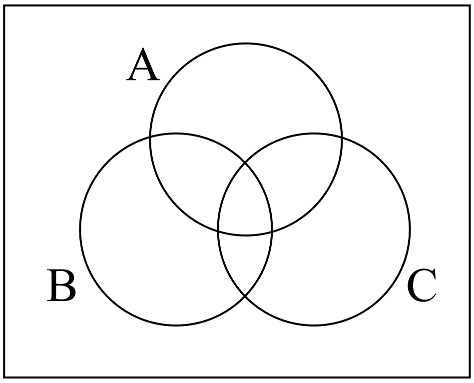


Eg 12: Let A = {a, b, c, e, f}, B = {b, e, f, r, s} and
C = {a, t, u, v}. Find $A \cap B$, $A \cap C$ and $B \cap C$.

Eg 13:



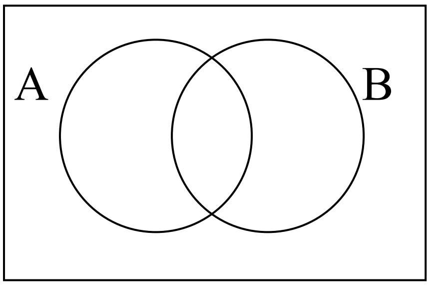
$A \cup B \cup C$



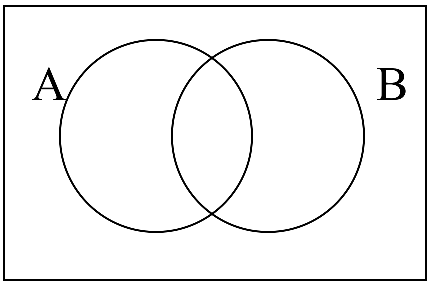
$A \cap B \cap C$

Eg 14: Let A = {1, 2, 3, 4, 5, 7}, B = {1, 3, 8, 9} and
C = {1, 3, 6, 8}. Find $A \cap B \cap C$.

3) **A – B** = $\{x \mid x \in A \text{ and } x \notin B\}$



A – B

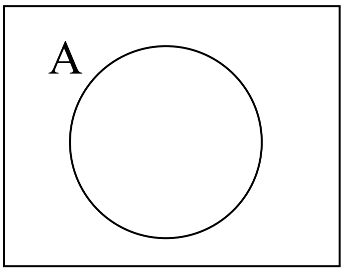


B – A

Eg 15: Let A = {a, b, c} and B = {b, c, d, e}. Find A – B, B – A.

4) **Complement set**

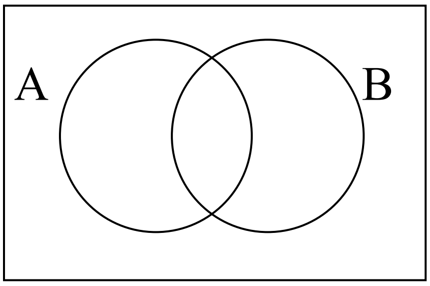
- If U is a Universal set containing A, then U – A is called the complement of A and is denoted by $\overline{A}$ or A' or A^c.
- $A' = \{x \mid x \notin A\}$



Eg 16: Let $A = \{x \mid x \text{ is an integer and } x \leq 4\}$ and U = Z.

A' =

5) **A ⊕ B** = (A – B) ∪ (B – A)



Eg 17: Let A = {a, b, c, d} and B = {a, c, e, f, g}. Then
A ⊕ B =

<u>Laws of Algebra of Sets</u>

1) Idempotent
a) $A \cup A = A$ b) $A \cap A = A$

2) Associative
a) $(A \cup B) \cup C = A \cup (B \cup C)$
b) $(A \cap B) \cap C = A \cap (B \cap C)$

3) Commutative
a) $A \cup B = B \cup A$
b) $A \cap B = B \cap A$

4) Distributive
a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

5) Identity

a) $A \cup \varnothing = A$ b) $A \cap U = A$

c) $A \cup U = U$ d) $A \cap \varnothing = \varnothing$

6) Involution

$(A^c)^c = A$ or $\overline{(\overline{A})} = A$

7) Complement

a) $A \cup A^c = U$ b) $A \cap A^c = \varnothing$

c) $\overline{U} = \varnothing$ d) $\overline{\phi} = U$

8) De Morgan's

a) $\left(\overline{A \cup B}\right) = \overline{A} \cap \overline{B}$ b) $\left(\overline{A \cap B}\right) = \overline{A} \cup \overline{B}$

- Duality Principal
  - In each of the laws of algebra of sets, if $\cup$ and $\cap$ are interchanged throughout and $\varnothing$ and U are interchanged throughout, the laws remain valid.

Eg 18: $(A \cap B) \cup (A \cap B^c)$ and write the dual of the result.

* In duality principle, $\cup \to \cap$, $\cap \to \cup$

$U \to \varnothing$, $\varnothing \to U$

Eg 19:

<u>Set equation</u> <u>Dual</u>

1) $(U\cap A)\cup(B\cap A)=A$

2) $(A\cap U)\cap(\varnothing\cap A^c)=\varnothing$

3) $(A\cap U)\cup(B\cap A)=A$

4) $A^c\cup B^c\cup C^c=(A\cap B\cap C)^c$

5) $(A^c\cap B^c)^c = A\cup B$

TM: If two sets A and B are disjoint, $A\cap B=\varnothing$

$$|A\cup B|=|A|+|B|$$

TM: If A and B are finite sets, then

$$|A\cup B|=|A|+|B|-|A\cap B|$$

TM: If A, B, C are finite sets, then

$$|A\cup B\cup C|=|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$$

Eg 20: Let A = {a, b, c, d, e} and B = {c, e, f, h, k, m}.

Verify $|A \cup B| = |A| + |B| - |A \cap B|$.

Eg 21: Let A = {a, b, c, d, e}, B = {a, b, e, g, h} and C = {b, d, e, g, h, k, m, n}. Verify

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Eg 22: A survey of movie lovers produced the following results: 30 like war movies, 42 like love stories, 55 like comedies, 4 like war and love stories, 20 like war movies and comedies, 28 like love stories and comedies, 2 like all three and 22 like none of the three.

(i) How many people were surveyed?
(ii) How many like war movies only?
(iii) How many like comedies only?
(iv) How many like love stories only?

Eg 23: A survey has been on methods of commuter travel. Each respondent was asked to check BUS, TRAIN or AUTOMOBILE as a major of traveling to work. More than one answer was permitted. The results reported were as follows: BUS, 30 people; TRAIN, 35 people; AUTOMOBILE, 100 people; BUS and TRAIN, 15 people; BUS and AUTOMOBILE, 15 people; TRAIN and AUTOMOBILE, 20 people; and all three methods, 5 people. How many people completed a survey form?

Mathematical Induction

is a mathematical method of proof.

If P(n) is a statement that is true for n (usually ≥ 0), then it can be proved as follows:

Step 1: Show that P(n) is true for some initial value n, usually 1.

Step 2: Assume that P(k) is true, for some $k \geq n$.

Step 3: Prove that P(k +1) is also true by using the assumption in step 2.

Eg 24: Show that

$$1+2+3+4+...+n=\frac{n(n+1)}{2}, n\geq 1$$

Eg 25: Show that

$$1^2 + 2^2 + ... + n^2 = \frac{n}{6}(n+1)(2n+1), \quad n \geq 1$$

Eg 26: Show that $3 \mid (n^3 - n)$, $\forall n \geq 1, n \in Z$

Sequences

- A sequence is simply a list of objects arranged in a definite order; a first element, second element, third element and so on.

Eg 27: The sequence 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1 is a finite sequence with repeated items.

Eg 28: The list 3, 8, 13, 18, 23,… is an infinite sequence.

Eg 29: 1, 4, 9, 16, 25,… is an infinite sequence with the list of the squares of all positive integers.

Eg 30: The recursive formula $c_1 = 5$, $c_n = 2c_{n-1}$, $2 \leq n \leq 6$, defines the finite sequence 5, 10, 20, 40, 80, 160.

➢ Characteristic Function

$$f_A(x) = \begin{cases} 1 & if \quad x \in A \\ 0 & if \quad x \notin A \end{cases}$$

Eg 31: U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
A = {2, 4, 6, 8, 10}
B = {1, 3, 5, 7, 9}
C = {2, 3, 5, 7}

x	1	2	3	4	5	6	7	8	9	10
$f_A(x)$										
$f_B(x)$										
$f_C(x)$										
$f_A(x).f_C(x)$										
$f_{A\cap C}(x)$										
$f_{A\cup C}(x)$										
$f_A(x)+f_C(x)-f_{A\cap C}(x)$										

Note that

$$f_{A\cup B}(x) = \begin{cases} 0 & iff \quad f_A(x)=0, f_B(x)=0 \\ 1 & \qquad otherwise \end{cases}$$

➢ Set Operations in Computers

1) Union

$$f_{A\cup B}(x) = \begin{cases} 0 & iff \quad f_A(x)=0, f_B(x)=0 \\ 1 & \qquad otherwise \end{cases}$$

2) Intersection

$$f_{A\cap B}(x)= \begin{cases} 1 & iff \quad f_A(x)=1, f_B(x)=1 \\ 0 & otherwise \end{cases}$$

3) Set difference: A – B is obtained by A ∩ B’

$$f_{A-B}(x)= \begin{cases} 1 & iff \quad f_A(x)=1, f_B(x)=0 \\ 0 & otherwise \end{cases}$$

Eg 32: Let U = {a, b, c, d, e, f, g, h}, A= {a, b, f}, B = {c, d, e, f, g}, C = {c, e, g, h}
Represent each of the sets A, B, C and U by bit strings. Find the bit string representation for

a) A ∪ B b) A ∩ C
c) B – A d) C’

Division in the integers

- For integers, n and q, if $\frac{m}{n} = q$, then we say "n divides m" or "m is divisible by n" or "m is a multiple of n" or "n is a divisor of m".

Eg 33: $\frac{4}{2} = 2$ $\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

- If n divides m $\Rightarrow$ $n \mid m$
- If n does not divides m $\Rightarrow$ $n \nmid m$

Eg 34: $2 \mid 4$ $\quad$ $5 \mid 10$ $\quad$ $3 \nmid 7$

- If $n \mid m$, then m = nq + r, where r = 0
- If $n \nmid m$, then m = nq + r where 0 < r < n.

Eg 35: $3 \mid 6 \Rightarrow$

$3 \nmid 7 \Rightarrow$

$4 \nmid 15 \Rightarrow$

TM: If $n \neq 0$ and m are non-negative integers, then m = nq + r for some non-negative integers q and r with $0 \leq r \leq n$. Moreover, there is just one way to do this.

TM: Let a, b, c are integers.

a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
b) If $a \mid b$ and $a \mid c$, where $b > c$, then $a \mid (b - c)$.
c) If $a \mid b$ or $a \mid c$, then $a \mid bc$.
d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Defn: A +ve integer $p > 1$ is called prime if it is only divisible by p and 1.

Eg 36: Prime numbers: 2, 3, 5, 7, 11, 13, …

8 is divisible by

TM: Every +ve integer $n > 1$ can be written uniquely as

$$n = p_1^{k_1} p_2^{k_2} \dots p_s^{k_s}$$

where $p_1 < p_2 < \dots < p_s$ are distinct prime, s and k are all +ve integers.

Eg 37: 10 =
25 =
70 =
36 =

Defn: If a, b, q are +ve integers such that $q \mid a$ and $q \mid b$, then q is a common divisor of a and b.

20:
24:
Common divisor are

Defn: If d is the largest integer that divides both a and b.
d is called greatest common divisor of a and b,
denoted by gcd (a, b) = d

Eg 38: gcd (20, 24) =
gcd (4, 10) =
gcd (12, 30) =

TM: If d = gcd (a, b), then

i) d = au + bv for some integers u and v (not both +ve)
ii) if c is another common divisor of a and b, then $c \mid d$.

Eg 39: gcd (20, 24) =4
$\therefore$ 4 = 20(-1) + 24(1)

gcd (12, 30) =

Defn: If gcd (a,b) = 1, then we say a and b are relatively prime.

Eg 40: 31 and 15 are relatively prime.

Now, if d = gcd (a, b), how to find d, u and v in
d = au + bv?

* If $a \mid b$, gcd (a,b) =

<u>Euclidean Algorithm</u>

Let $a > b$, then
$$a = bq_1 + r_1, \qquad 0 \le r_1 < b$$
$$b = r_1 q_2 + r_2, \qquad 0 \le r_2 < r_1$$
$$r_1 = r_2 q_3 + r_3, \qquad 0 \le r_3 < r_2$$
$$\vdots$$
$$r_{n-1} = r_n q_{n+1} + r_{n+1}, \qquad 0 \le r_{n+1} < r_{n-1}$$

As $b > r_1 > r_2 > \ldots > r_{n+1} \ge 0$

Eventually, $r_{n+1} = 0$

Let $d = \gcd(a, b)$

Since $d \mid a$, $d \mid b$ then $d \mid r_1$, so $d \mid r_i$, $\forall i$.

If $d = \gcd(a, b)$, $d = \gcd(b, r_1)$
$$= \gcd(r_1, r_2)$$
$$\vdots$$
$$= \gcd(r_n, r_{n+1})$$

Since $r_{n+1} = 0$, $\gcd(r_n, r_{n+1}) = r_n$

$\frac{m}{n} \in Z$, $\quad n \mid m \qquad\qquad \frac{m}{n} \notin Z$, $\quad n \nmid m$

$m = nq + r, \qquad 0 \le r < n$

If $a \mid m$, $a \mid n$, a is a common divisor of m and n.

If d is the largest common divisor,

$d = \gcd(m,n)$

$d = mu + nv, \qquad u,v \in Z$

$\gcd(2, 4) =$

$2 = 2(\ \ ) + 4(\ \ ) = 2(\ \ ) + 4(\ \ )$

* If $m \mid n$, gcd (m, n) = m

Eg 41: Find

a) gcd (10, 8)   b) gcd (100, 52)

Eg 42: Find the gcd of the integers a and b and write
$d = au + bv$.

a) a = 60, b = 100
b) a = 45, b = 33

Given 7 and 4 Given 10 and 1

Defn: Let a, b, n be integers with $n > 0$, then a is congruent to b modulo n, denoted $a \equiv b \pmod{n}$. If "$n \mid (a - b)$ or $n \mid (b - a)$" or both a and b give the same remainder when divided by n.

Eg 43: $3 \mid (7 - 4)$ $3 \mid (5 - 2)$

$\Rightarrow 7 \equiv 4 \pmod{3}$ $\Rightarrow$

$\Rightarrow 4 \equiv 7 \pmod{3}$ $\Rightarrow$

If $a \equiv b \pmod{n}$, $n \mid (a - b)$

TM: If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Boolean Matrices

- A matrix whose entries are 0 or 1.

Let A = [a_{ij}], B = [b_{ij}], C = [c_{ij}]

1) The joint of A and B,
   $A \vee B = C = [c_{ij}]$
   where $c_{ij} = \begin{cases} 1, & a_{ij} = 1 \quad or \quad b_{ij} = 1 \\ 0, & a_{ij} = 0 \quad and \quad b_{ij} = 0 \end{cases}$

2) The meet of A and B,
   $A \wedge B = [c_{ij}]$
   where $c_{ij} = \begin{cases} 1, & a_{ij} = 1 \quad and \quad b_{ij} = 1 \\ 0, & a_{ij} = 0 \quad or \quad b_{ij} = 0 \end{cases}$

3) Boolean product,
   $A \odot B = [c_{ij}]$

$$c_{ij} = \begin{cases} 1, & \text{if } \exists k \text{ such that } a_{ik} = 1 \text{ and } b_{kj} = 1 \\ 0, & \text{otherwise} \end{cases}$$

* no. of columns of A = no. of rows of B

Eg 44: $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

$A \vee B =$

$A \wedge B =$

Eg 45: $A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$

$A \odot B =$

Eg 46: $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A \odot B =$

Chapter 2 Logic of Compound Statements

Logic

- Deals with the methods of reasoning. It provides rules and techniques for determining whether a given argument is valid.

Statements

- A declarative sentence that is either true or false but not both.
- The truthfulness or falsity of a statement is called its truth value.
- True represented by T or 1.
- False represented by F or 0.

Eg 1: Identify the statements.

a) The earth is round.
b) $2 + 3 = 6$.
c) Ms Lee is fat and short.
d) Segamat is located in Seremban.
e) Take two aspirins.
f) The sun will come out tomorrow.
g) What time is it ?
h) $x + 1 = 2$.
i) Read this carefully.
j) $x + y = z$.

- Letters like p, q, r, s or P, Q, R, S are used to denote prepositional variables.
- A true statement has truth value T.
- A false statement has truth value F.
- If P is true, then ¬P is false.
- If P is false, then ¬P is true.

Eg 2: R : The earth is round has truth value
S : 2 + 3 = 6 has truth value

- Statement can be combined by logical connectives
- " **and** or **conjunction**" (Symbol ∧),
- " **or** or **disjunction**" (Symbol ∨),
- " **not** or **negation**" (Symbol ¬ or ~),

- The truth value depends on the types of connectives used.

<u>Truth Table</u>

- a table giving the truth values of a compound statement in terms of the truth values of its component parts.
- Each component statement has 2 possible truth values, T and F.
- If there are n component statements in the compound statements, then the number of rows in its truth table is 2^n.

Types of Connectives

1) Conjunction (and, $\wedge$)

Let P and Q be two simple statements, the conjunction of P and Q is the compound statement P and Q
($P \wedge Q$). <u>It is true when both P and Q are true.</u>

Eg 3: $R \wedge S$ means the conjunction of R and S is "The earth is round and 2 + 3 = 6".
Truth table for $P \wedge Q$

P	Q	$P \wedge Q$
T	T	
T	F	
F	T	
F	F	

2) Disjunction (or, $\vee$)
The compound statement is true if a t least one of P or Q is true.

Eg 4: $R \vee S$ means the disjunction of R and S is " The earth is round or 2 + 3 = 6"
Truth table for $P \vee Q$

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

3) Negation (not, $\neg$, ~)
~P is true when P is false.

Eg 5: ~S means the negation of S is " It is not the case that 2 + 3 = 6" or $2 + 3 \neq 6$.
Truth table for negation

P	~P
T	
F	

4) Implication (if…then…, $\rightarrow$)
The implication $p \rightarrow q$ is the proposition that is false when p is true and q is false. In this implication p is called the hypothesis (premise) and q is called the conclusions (consequence).

Truth table for $P \rightarrow Q = \sim P \vee Q$

P	Q	$P \rightarrow Q$	~P	$\sim P \vee Q$
T	T			
T	F			
F	T			
F	F			

Eg 6: "If I am your lecturer, then you all are my students" is obviously true.

"If I am your lecturer, then you all are my children" is false since the conclusion is false.

"If I am your father, then $2 + 3 = 5$" is true since the conclusion is true.

"If I am your father, then you all are my children" is definitely true though both hypothesis and conclusion are false.

Eg 7: "If you come on Sunday, I will employ you" is false only if you come on Sunday but I don't employ you.

5) Biconditional Statement (if and only if, iff, $\leftrightarrow$)

If $P\rightarrow Q$ and $Q\rightarrow P$, we write $P\leftrightarrow Q$ and read as P if and only if (iff) Q.

It is true only when both P and Q have the same truth values.

$P\leftrightarrow Q = (P\rightarrow Q) \wedge (Q\rightarrow P)$

Truth table for $P\leftrightarrow Q$

P	Q	$P\leftrightarrow Q$	$(P\rightarrow Q)\wedge(Q\rightarrow P)$
T	T		
T	F		
F	T		
F	F		

Tautology

- A proposition that is always true whatever the truth values of the component statements.
- The last column of the truth table consists of T only.

Eg 8: $P \vee \sim P$

P	$P \vee \sim P$
T	
F	

Contradiction

- A proposition that is always false whatever the truth values of the component statements.
- The last column of the truth table consists of F only.

Eg 9: $P \wedge \sim P$

P	$P \wedge \sim P$
T	
F	

Contingency

- A proposition that can be either true (T) or false (F), depending on the truth values of its component statements.

Eg 10: $\sim(P \wedge \sim Q)$

P	Q	$\sim(P \wedge \sim Q)$
T	T	
T	F	
F	T	
F	F	

- The order of evaluating the truth values of an expression is ( ), $\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$.

Eg 11: Show that $(P\rightarrow Q) \wedge (Q\rightarrow R) \rightarrow (P\rightarrow R)$ is tautology.

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	$A \cap B$	$(A \cap B) \rightarrow C$

Eg 12: Verify if the following are tautology, contradiction or contingency.

1) $\sim(P \wedge Q) \rightarrow P$ 2) $P \wedge \sim(P \vee Q)$

3) $\sim(P \rightarrow Q) \rightarrow P$

Converse, Inverse, Contrapositive, Negation

- The converse of $P \to Q$ is the implication $Q \to P$.
- The inverse of $P \to Q$ is the implication $\sim P \to \sim Q$.
- The contrapositive of $P \to Q$ is the implication $\sim Q \to \sim P$.
- The negation of $P \to Q$ is $P \wedge \sim Q$.

Eg 13: Give the converse, inverse, contrapositive and negation of the implication
"If it is raining, then I get wet."

Eg 14: p = I am hardworking.
q = I am successful.

$p \to q$:

$q \to p$:

$\sim p \to \sim q$:

$\sim q \to \sim p$:

$\sim(p \to q)$:

Logical Equivalence ($\equiv$ or $\Leftrightarrow$)

- Two statements are considered to be equivalent if they have the same truth value for any combination of truth values of their component statements.
- The last column of their truth table is identical. This means the biconditional statement of two logically equivalent.

Eg 15: Show that $P \wedge (Q \vee R)$ and $(P \wedge Q) \vee (P \wedge R)$ are logically equivalent.

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \vee (P \wedge R)$

Below are some important equivalence and their names.

1) $P \wedge T \Leftrightarrow P$ $P \vee F \Leftrightarrow P$	Identity laws
2) $P \vee T \Leftrightarrow T$ $P \wedge F \Leftrightarrow F$	Domination laws
3) $P \vee P \Leftrightarrow P$ $P \wedge P \Leftrightarrow P$	Idempotent laws
4) $\sim(\sim P) \Leftrightarrow P$	Double negation
5) $P \vee Q \Leftrightarrow Q \vee P$ $P \wedge Q \Leftrightarrow Q \wedge P$	Commutative laws
6) $P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$ $P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$	Associative laws
7) $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$ $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$	Distributive laws
8) $\sim(P \wedge Q) \Leftrightarrow \sim P \vee \sim Q$ $\sim(P \vee Q) \Leftrightarrow \sim P \wedge \sim Q$	De Morgan's law
9) $P \to Q \Leftrightarrow \sim P \wedge Q$ $P \wedge \sim P \Leftrightarrow F$ $P \vee \sim P \Leftrightarrow T$	

Eg 16: Show that $\sim(P \vee (\sim P \wedge Q))$ are logically equivalent to $\sim P \wedge \sim Q$ using laws of logic and truth table.

Eg 17: Show that $(P \wedge Q) \rightarrow (P \vee Q)$ is a tautology using laws of logic and truth table.

Normal Forms

- It is useful to have standard forms for expressions, because this makes the identification and comparison of two expressions easier.

- The standard forms for logical expression is called Normal Forms.

There are two types :

a) Disjunctive Normal Forms
   - A logical expression, which is written as a disjunction in which all terms are conjunctive of literals.

Eg 18: 1) $(P \wedge Q) \vee (P \wedge \sim Q)$
2) $P \vee (Q \wedge R)$
3) $\sim P \vee R$

Eg 19: Is the expression $\sim(P \wedge Q) \vee R$ in disjunctive normal form? Why?

b) Conjunctive Normal Forms

- A logical expression, which is written as a conjunction in which all terms are disjunctions of lateral.

Eg 20: 1) $P \wedge (Q \vee R)$

2) $P \wedge F$

3) Is $P \wedge (R \vee (P \wedge Q))$ in conjunctive normal form? Why?

<u>Obtain Disjunctive/ Conjunctive Normal Forms through algebraic manipulation</u>

Three steps are require:

1) Remove all $\rightarrow$ and $\leftrightarrow$.

Eg) $P \rightarrow Q \equiv {\sim}P \vee Q$

$P \leftrightarrow Q \equiv ({\sim}P \vee Q) \wedge ({\sim}Q \vee P)$

2) If the expression in question contains any negated compound sub expressions, either remove the negation by using the double negation law or De Morgan's laws to reduce the scope of the negation.

3) Once an expression with no negated compound sub expression is found, use the distributive laws to reduce the scope to $\wedge$ and $\vee$.

$P \vee (Q \wedge R) \equiv$

$P \wedge (Q \vee R) \equiv$

$(P \wedge Q) \vee R \equiv$

$(P \vee Q) \wedge R \equiv$

Eg21: Convert the expression $\sim((P \vee \sim Q) \wedge \sim R)$ in to normal form.

Eg 22: Obtain disjunctive and conjunctive normal form of $P \wedge (P \rightarrow Q)$.

Principle Disjunctive Normal Forms

- The disjunctive normal form and the conjunctive normal form of a given expression are not unique, such that there may be several normal forms which are equivalent to each other.. For eg, $P \vee (Q \wedge R)$ is already in disjunctive normal form. We may also express it in another disjunctive normal form as :

  $P \vee (Q \wedge R) \equiv$

- Comparison of expression would be easier if they can be converted into some standard form that does not have any variations, preferable that is unique for each expression.

- We define a min term or Boolean conjunction of several simple statement variables P, Q, R… as product of P,Q, R… or their negations, formed in such a way that each variable appears exactly one either as itself or its negation.

Eg 23: For 2 variables P and Q, the min terms are :

Eg 24: For 3 variables P, Q and R, the min terms are :

- Any expression that can be obtained by commuting the factors in the expressions above is not included in the list, as it would be equivalent to one of the min terms above.

- The truth table of these min terms are given below :

P	Q	$P \wedge Q$	$P \wedge \sim Q$	$\sim P \wedge Q$	$\sim P \wedge \sim Q$

From the truth table,

1) Each min term has the truth value T for exactly one combination of the truth values of P and Q.
2) No two min terms are equivalent.

Eg 25: Construct the truth tables for $P \to Q$, $P \vee Q$ and $\sim(P \wedge Q)$ and obtain the PDNF of these expressions.

P	Q	$P \to Q$	$P \vee Q$	$P \wedge Q$	$\sim(P \wedge Q)$

Principle Conjunctive Normal Form

- For a given number of variables, a max term consists of disjunctions in which each variable or its negation, but not both, appears only once. The max terms are the duals of min terms.
- Each of the max term has the truth value F for exactly one combination of the truth values of the variables. Different max terms have the truth value F for different combinations of the truth values of the variables.
- For a given expression, an equivalent expression consisting of conjunctions of the max terms only is called its principle conjunctive normal form (PCNF) or the product-of-sums canonical form.
- In a PCNF, the max terms included correspond to the F values in the truth table of that expression. It is written down by including the variable if its truth value is F and its negation if the value is T.
- Every logical expression which is not a tautology has an equivalent PCNF, which is unique except for the

rearrangement of the factors in the max terms as well as in their disjunctions.

PCNF max terms :

P, Q :

P, Q, R :

Eg 26: Find the PCNF of the expression in last example.

- ❖ If the PDNF (or PCNF) of a given expression. A is known, then the PDNF (or PCNF) of ~A will consists of the disjunction (or conjunction) of the remaining min terms (or max terms) which do not appear in the PDNF (or PCNF) of A.
- ❖ From A ≡ ~~A, we can obtain the PCNF (or PDNF) of A by repeated applications of De Morgan's laws to the PDNF (or PCNF) or ~A.

Eg 27: Let A represent $(\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$. Obtain the PCNF and PDNF of A and ~A.

P	Q	R	$\sim P$	$(\sim P \rightarrow R)$	$(Q \leftrightarrow P)$	$(\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$

Eg 28: Repeat the above example without using truth table.

$$A \equiv (\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$$

Predicate Logic

- Consider the statement.
  - Lim Yoke Hong is a good student from TARC.
    The subject is
    The predicate is

- In logic, if we let P stands for "is a good student from TARC" and Q stand for "is a good student from", then both P and Q are predicate symbols and
  - "x is a good student from TARC", and
  - "x is a good student from y".

- P(x) and Q(x, y) are not statements as their truth values cannot be determined. They are called statement functions.

Defn: A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables.

The domain of a predicate variable is the set of all values that may be substituted in place of the variable.

Eg 29: For P(x) above, the domain is the set $\{x \mid x \text{ is a name}\}$.

Defn: If $P(x)$ is a predicate and x has domain D, the truth set of $P(x)$ is the set of all elements of D that make $P(x)$ true when substituted for x, denoted $\{x \in D \mid P(x)\}$.

Eg 30: Let $P(x)$ be "is a multiple of 2". Then the truth set of $P(x)$ is {all even integers}.

Eg 31: Let $Q(x)$ be "x is a factor of 8". Then the truth set of $Q(x)$ is $\{1, 2, 4, 8\}$.

<u>Notation</u>

- Let $P(x)$ and $Q(x)$ be predicates and suppose the common domain of x is D.

$P(x) \Rightarrow Q(x)$ means that every element in the truth set of $P(x)$ is in the truth set of $Q(x)$.

$P(x) \Leftrightarrow Q(x)$ means that $P(x)$ and $Q(x)$ have identical truth sets.

Eg 32: Let P(x) be "x is a factor of 8"
Q(x) be "x is a factor of 4"
R(x) be "x<5 and $x \neq 3$"
Domain of x is Z^+.
Use the $\Rightarrow$ and $\Leftrightarrow$ symbols to indicate the relationships among P(x), Q(x) and R(x).

Quantifiers: $\forall$ and $\exists$

- Expression containing variable can be made into statements by adding quantifiers.

- Quantifiers are words that refer to quantifiers such as "some" or "all" that tells for how many elements a given predicate is true.

$\forall$ denotes: "for all" and is called the universal quantifier.

Eg 33: a) "All human beings are mortal" can be expressed

b) $\forall x, x > 0$ means

Defn: Let Q(x) be a predicate and D be the domain of x. A universal statement is a statement of the form "$\forall x \in$ D, Q(x).It is true iff Q(x) is true for all x in D and false iff Q(x) is false for at least one x in D. A value for x for which Q(x) is false is called the counterexample to the universal statement.

Eg 34: a) Let D = {1, 2, 3, 4, 5} and consider the statement

$$\forall x \in \text{D}, x^2 \geq x.$$

Show that this statement is true.

b) Consider the statement

$$\forall x \in \Re, x^2 \geq x.$$

Find a counterexample to show that this statement is false.

- To show Q(x) is true by checking through all x is called the method of exhaustion (to exhaust all possibilities before you exhaust yourself).

- The symbol ∃ denotes "there exist" or "there is at least one" or "some" and is called the existential quantifier.

Eg 35: There is a student in Math 140 can be expressed as

Eg 36: a) Consider the statement:

$$\exists\, m \in Z \text{ such that } m^2 = m.$$

Show that this statement is true.

b) Let E = {5, 6, 7, 8, 9, 10} and consider the statement:

$$\exists\, m \in E \text{ such that } m^2 = m.$$

Show that this statement is false.

Eg 37: Rewrite the following formal statements in a variety of equivalent but more informal ways. Do not use the symbol $\forall$ or $\exists$.

a) $\forall x \in \Re, x^2 \geq 0$.

b) $\forall x \in \Re, x^2 \neq -1$.

c) $\exists$ m $\in$ Z such that $m^2 = m$.

Eg 38: Rewrite each of the following statements formally. Use quantifiers and variables.

a) All triangles have three sides.

b) No dogs have wings.

c) Some programs are structured.

Universal Conditional Statements

$\forall x$, if $P(x)$ then $Q(x)$

Eg 39: Rewrite the following formal statement in a variety of informal ways. Do not use quantifiers or variables.

$\forall x \in \Re$ if $x > 2$ then $x^2 > 4$

Equivalent Forms of Universal and Existential Statements

"$\forall x \in U$, if $P(x)$ then $Q(x)$" is equivalent to "$\forall x \in D$, $Q(x)$" if D is the domain consists of all values of the variable x that make $P(x)$ true.

Eg 40: The following statements are equivalent

$\exists$ a number n such that n is prime and n is even.

and $\exists$ a prime number n such that n is even.

Negation of Quantified Statements

- Give the negation of "All mathematician wear glasses".
  1) One or more mathematician does not wear glasses.
  2) Some mathematicians do not wear glasses.

- The negation of a statement of the form "$\forall x$ in D, Q(x)" is logically equivalent to a statement of the form "$\exists x$ in D such that ~Q(x).

$$\sim(\forall x, Q(x)) \equiv \exists x, \sim Q(x)$$

- The negation of a statement of the form "$\exists x$ in D such that Q(x)", is logically equivalent to a statement of the form "$\forall x$ in D, ~Q(x)".

$$\sim(\exists x, Q(x)) \equiv \forall x, \sim Q(x)$$

Eg 41: Negate the following statements.

1) $\forall$ primes p, p is odd.
2) No politicians are honest.
3) There is a patriotic citizen that supports the ruling party.
4) Some computer hackers are 40 or below.

Using if-then statement, we have
$\sim(\forall x$, if $P(x)$ then $Q(x)) \equiv \exists x$ such that $P(x)$ and $\sim Q(x)$

An expression may be true in a certain domain of interpretation and false in some other domains.

Statement	When is it true?	When is it false?
$\exists x, P(x)$	For some (at least one) a in the universe, P(a) is true.	For every a in the universe, P(a) is false.
$\forall x, P(x)$	For every a in the universe, P(a) is true.	There is at least one a in the universe for which P(a) is false.
$\exists x, \sim P(x)$	For at least one a in the universe, ~P(a) is true or P(a) is false.	For all a in the universe, ~P(a) is false or P(a) is true.
$\forall x, \sim P(x)$	For all a in the universe, ~P(a) is true or P(a) is false.	For at least one a in the universe, ~P(a) is false, or P(a) is true.

Eg 42: What is the truth value of $\forall x, x \geq 0$ if the domain of interpretation consists of:

a) the price of software in a computer shop.

b) All the integers.

Eg 43: Suppose $Q(x)$ represents "x is less than 5". Determine the truth value of the statements. $\forall x, \sim Q(x)$ and $\exists x, Q(x)$ if the universe of discourse is given by the sets:

a) $\{-1, 2, 4\}$

b) $\{3, -2, 5, 7\}$

c) $\{15, 20, 30\}$

Scope of Quantifier

- Expressions can be built by using quantifiers together with the usual connectives. Parentheses and brackets help to identify the scope of quantifier, i.e. the section of the expression to which the quantifier applies.

Eg 44: $(\forall x)[p(x) \wedge q(x)]$

Eg 45: $(\forall x)((\exists x)\ [p(x,y) \wedge q(x,y)])$

- Free Variable: a variable that has occurrence not within the scope of a quantifier involving that variable.

Eg 46: $(\forall x)((\exists y)p(x,y) \wedge q(x,y))$

The second occurrence of y is outside the scope of $(\exists y)$, so y is a free variable.

- An expression having any free variable does not qualify to be a statement as its truth value cannot be determined.

Chapter 3 Relations and Digraphs

Products Sets

- An ordered pair (a,b) is a listing of the objects a and b in a prescribed order.

$$A\times B = \{(a,b) \mid a\in A \text{ and } b\in B\}$$

Eg 1: Let A ={1, 2, 3} and B = {r, s}

$A\times B =$

$B\times A =$

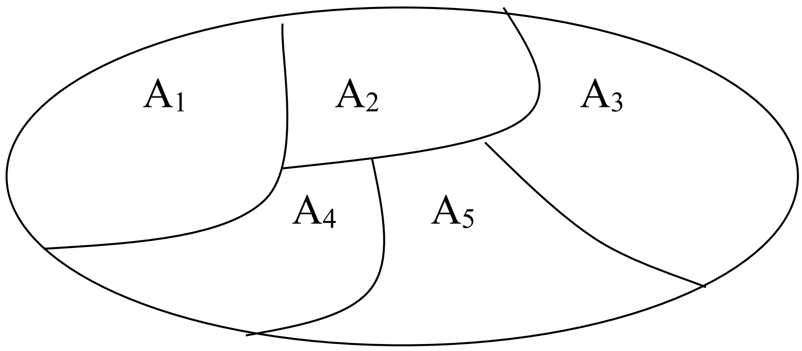
$A\times A =$

$A\times B\times B=$

- In general, $A\times B \neq B\times A$, but $|A\times B| = |B\times A|$.
- For finite sets, $A_1, A_2, A_3,\ldots, A_n$,
  the Cartesian products $A_1\times A_2\times A_3\times\ldots\times A_n$
  $=\{(a_1, a_2, a_3,\ldots, a_n): a_1\in A_1, a_2\in A_2,\ldots a_n\in A_n\}$

Partitions

- A partition or quotient set of a nonempty set A is a collection ρ of nonempty subsets of A such that
  1. Each element of A belongs to one of the sets in ρ.
  2. If A_1 and A_2 are distinct elements of ρ, then $A_1\cap A_2 = \varnothing$.



Eg 2: Let A = {a, b, c, d, e}. Which of the following is/are partitions of A?

i) $\rho = \{\{a,b\},\{c\},\{d\},\{e\}\}$

ii) $\rho = \{\{a,b\},\{c,d\}\}$

iii) $\rho = \{\{a\},\{b\},\{c\},\{d\}\}$

iv) $\rho = \{\{a,b\},\{c,d\},\{e\}\}$

v) $\rho = \{\{a\},\{b,d\},\{c\},\{d,e\}\}$

Eg 3: List all partitions of A = {1, 2, 3}.

Relations and Digraphs

- Let A and B be nonempty sets. A relation R from A to B is a subset of $A \times B$.
- If $R \subseteq A \times B$ and $(a,b) \in R$, we say that a is related to b by R and write as aRb.
- If a is not related to b by R, we write $a \not R b$.

Eg 4: Let A = {1, 2, 3} and B = {r, s}. Then R = {(1,r), (2,s), (3,r)}is a relation from A to B.

Eg 5: Let A and B be sets of real numbers. We define the following relation R (equals) from A to B.

aRb if and only if a = b

Eg 6: Let A = {1, 2, 3, 4, 5}. Define the following relation R (less than) on A.

aRb iff $a < b$

Eg 7: Let A = {1,2,3,4}. Define

$x\mathrm{R}y$ iff $x \mid y$

$x\mathrm{S}y$ iff $x = y$

Find R and S.

Domain of R

- Dom R = $\{x \in X \mid xRy \text{ for some } y \in Y\}$
  = all the x values of (x,y) in R

Range of R

Ran R = $\{y \in R \mid xRy \text{ for some } x \in X\}$

Eg 8: Let A = {1, 2, 4, 6}. Define aRb iff a | b.
Find the domain and range.

Eg 9: Let A = {1, 2, 4, 6}. Define $a \leq b$ iff $a \equiv b \pmod 2$.
Find the domain and range.

- Let R be a relation from A to B and $x \in A$, the R-relative set of x, R(x), is the set of all $y \in B$ which are related to x by R.

$$R(x) = \{y \in B \mid xRy\}$$

Eg 10: Let A = {a, b, c, d}.
R = {(a,a), (a,c), (b,c), (b,d), (c,a), (c,d), (d,b), (d,c)}
R(a) =
R(b) =
R(c) =
R(d) =

Eg 11: From Eg 9,

$S(1) =$

$S(2) =$

$S(4) =$

$S(6) =$

$A/S =$

TM: Let R be a relation from A to B, and let A_1 and A_2 be subsets of A. Then

a) If $A_1 \subseteq A_2$, then $R(A_1) \subseteq R(A_2)$.

b) $R(A_1 \cup A_2) = R(A_1) \cup R(A_2)$

c) $R(A_1 \cap A_2) \subseteq R(A_1) \cap R(A_2)$

Matrix of a relation

- For a matrix with m rows and n columns, $M_{m,n}$, the element in row I and column j is denoted by m_{ij}.
- If $A = \{a_1, a_2, \ldots, a_m\}$ and $B = \{b_1, b_2, \ldots, b_n\}$ are finite sets, R be a relation from A to B, we represent R by $m \times n$ matrix.

$$M_R = [m_{ij}]$$

$$m_{ij} = \begin{cases} 1 \ if \ (a_i, b_j) \in R \\ 0 \ if \ (a_i, b_j) \notin R \end{cases}$$

Eg 12: $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$

$R = \{(1,a), (1,c), (2,b), (2,d), (3,a), (3,c), (3,d)\}$

$S = \{(a,1), (a,3), (b,2), (c,3), (d,2)\}$

Find M_R and M_S.

Eg 13: A = {1, 3, 5}, B = {a, b, c, d}. R is from A to B

with $M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$. Find R.

Digraphs of a relation

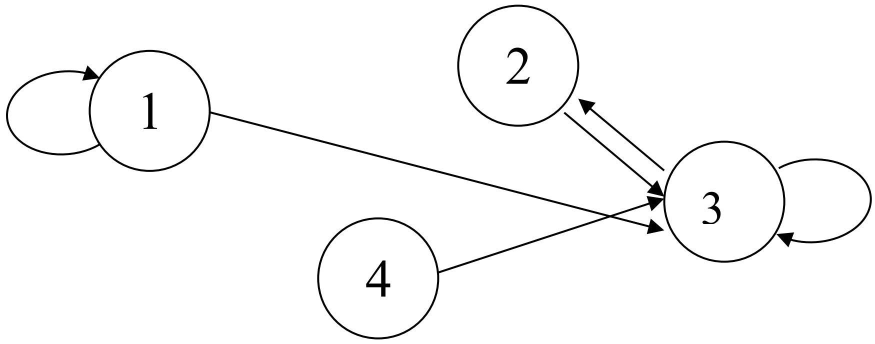
- If A is a finite set and R is a relation on A, we can also represent pictorially.
- Draw a small circle for each element of A and label the circle with the corresponding element of A. These circles are called vertices.
- Draw an arrow, called an edge from vertex a_i iff a_iRa_j.
- The resulting pictorial representation of R is called a directed graph or digraph of R.

Eg 14: Let A = {1, 2, 3, 4}

R = {(1,1), (1,2), (2,1), (2,2), (2,3), (2,4), (3,4), (4,1)}

Draw digraph of the relation.

Eg 15: Find the relation determined by

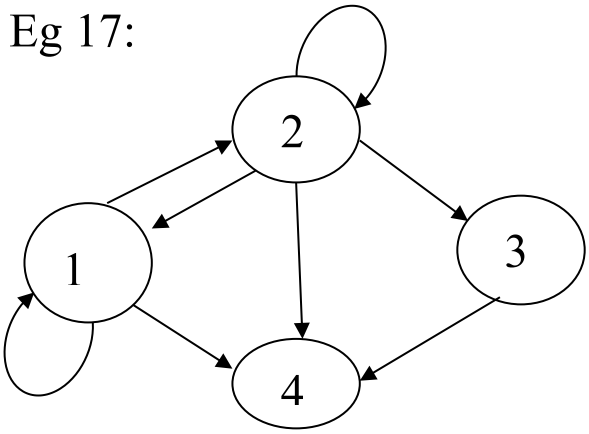


Eg 16: Let A = {a, b, c, d, e}. Given $M_R = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

Draw the digraph of R.

- If R is a relation on a set A and a∈A, then the **in-degree** of a is the number of b∈A such that (b,a)∈R. The **out-degree** of a is the number of b∈A such that (a,b)∈R.

Eg 17:



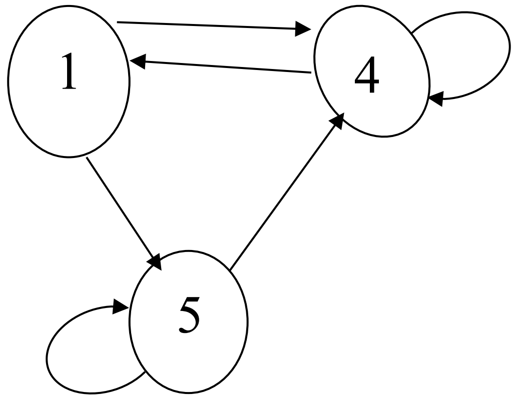
Vertex	1	2	3	4
In-degree				
Out-degree				

Eg 18: Let A = {a, b, c, d} and let R be the relation on A that has the matrix

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Construct the digraph of R, and list in-degrees and out-degrees of all vertices.

Eg 19: Let A = {1, 4, 5} and let R be given by the digraph below. Find M_R and R.



➢ If R is a relation on a set A, and B is a subset of A, the restriction of R to B is $R\cap(B\times B)$.

Eg 20: Let A = {a, b, c, d, e, f} and
R = {(a,a), (a,c), (b,c), (a,e), (b,e), (c,e)}
Let B = {a, b, c}. Then

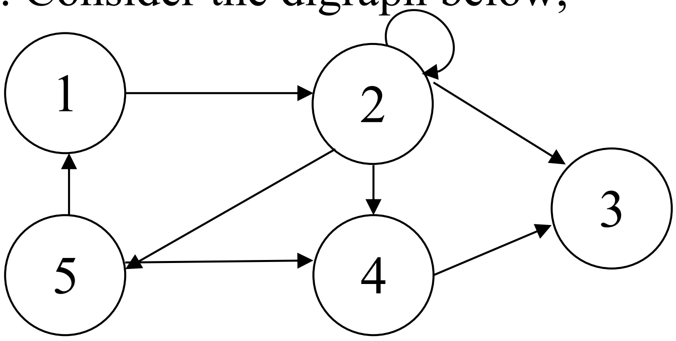
$B\times B =$

$R\cap(B\times B) =$

Paths in Relations and Digraphs

- Suppose that R is a relation on a set A. A path of length n in R from a to b is a finite sequence π: a, x_1, $x_2,\ldots,x_{n-1}$,b, such that aRx_1, $x_1Rx_2,\ldots,x_{n-1}$Rb.

Eg 21: Consider the digraph below,

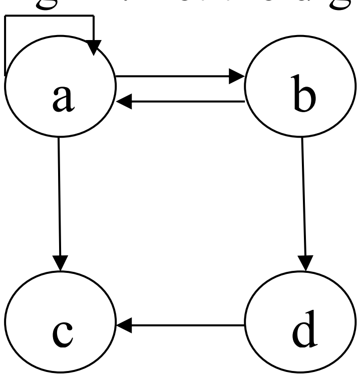


π_1 = 1, 2, 5, 4, 3 $\Rightarrow$
π_2 = 1, 2, 5, 1 $\Rightarrow$
π_3 = 2, 2 $\Rightarrow$

- A path that begins and ends at the same vertex is called a cycle.
- xR^ny means that there is a path of length n from x to y in R.
- $xR^\infty y$ means that there is some path in R from x to y.

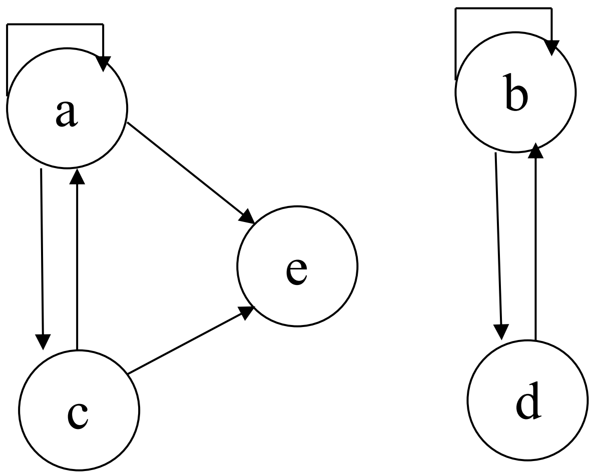
  $R^\infty \Rightarrow$ connectivity relation
  $R^\infty = R^1 \cup R^2 \cup R^3 \cup\ldots$

Eg 22: Let the digraph of R on A = {a, b, c, d} a below:



ac is a path of length ____
abac is a path of length ____
aabdc is a path of length ____
aa is a path of length _____
aba is a cycle of length _____
aa is a cycle of length______

Eg 23:

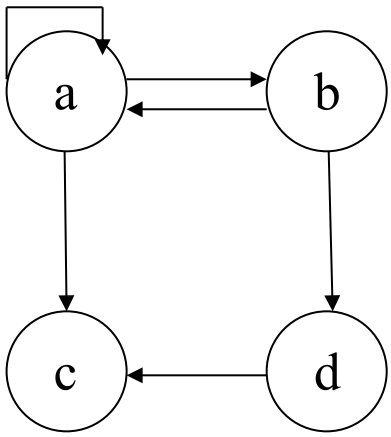


acae $\Rightarrow$

acaae $\Rightarrow$

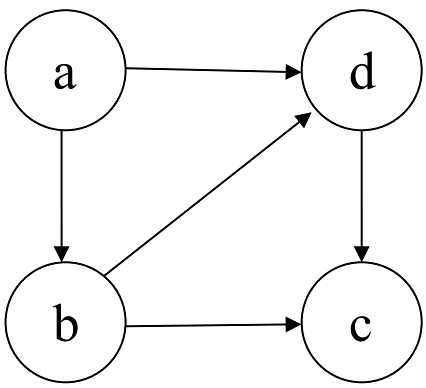
ae $\Rightarrow$

Eg 24:



$R^{\infty} =$

Eg 25:



$R^2 =$

$R^{\infty} =$

- In matrix, $M_{R^\infty} = M_R \vee M_{R^2} \vee M_{R^3} \vee \ldots$

TM: Let M_R be the relation matrix of R on A. Then

$$M_{R^2} = M_R \odot M_R$$

$$= (M_R)^2_\odot$$

Eg 26: Let R be a relation on A with

R = {(a,a), (a,c), (b,a), (b,d), (a,b), (d,c)}.

Find M_R and M_{R^2}.

- $M_{R^n} = M_R \odot M_{R^2} \odot M_{R^3} \ldots \odot M_{R^n}$

Eg 27: From the previous eg, find $M_{R^3}, M_{R^4}, M_{R^\infty}$.

- The reachability relation R^* of a relation R on a set A that has n elements is defined as follows:
  xR^*y means that $x = y$ or $xR^\infty y$.
  $$M_{R^*} = M_{R^\infty} \vee I_n$$ where I_n is the n×n identity matrix.

Eg 28: $M_{R^\infty} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$$M_{R^*} = M_{R^\infty} \vee I_4$$

$$=$$

- Let π_1: a, $x_1, x_2, \ldots, x_{n-1}$,b be a path in a relation R of length n from a to b and let π_2: b, $y_1, y_2, \ldots, y_{m-1}$,c be a path in R of length m from b to c. Then the composition of π_1 and π_2 is the path a, $x_1, x_2, \ldots, x_{n-1}$,b, $y_1, y_2, \ldots, y_{m-1}$,c of n + m, which is denoted by $\pi_2 o \pi_1$. This is a path from a to c.

Eg 29: π_1 = abcadb, π_2 = badefc

$\pi_2 \circ \pi_1 =$

Length of $\pi_2 \circ \pi_1$=sum of length π_1 & π_2

= length of π_2 + length of π_1

=

Properties of Relations

1) Reflexive

- A relation R on a set A is reflexive if $(a,a) \in R$ for all $a \in A$, that is aRa for $a \in A$.

2) Irreflexive

- A relation R on a set A is irreflexive if $a \not R a$ for every $a \in A$.

Eg 30: Let $\Delta = \{(a,a) \mid a \in A\}$.

Δ is the relation of equality on the set A.

Δ is reflexive, since $(a,a) \in \Delta$ for all $a \in A$.

Eg 31: Let $R = \{(a,b) \in A \times A \mid a \neq b\}$.

R is the relation of inequality on the set A.

R is irreflexive, since $(a,a) \notin R$ for all $a \in A$.

Eg 32: Let A = {1, 2, 3}.

R = {(1,1), (1,2)}

Eg 33: Let A be a nonempty set.

$R = \varnothing \subseteq A \times A$, the empty relation.

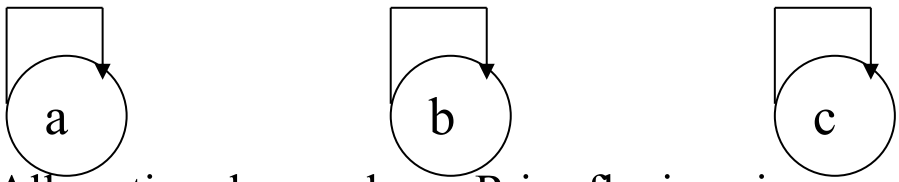
➢ In matrix, if R is reflexive, M_R has this form:

$$M_R = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$ - main diagonal are "all 1".

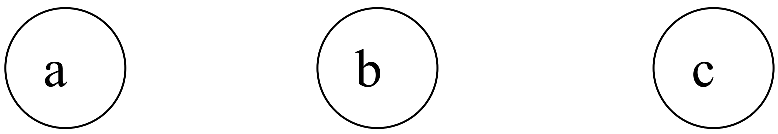
➢ If R is irreflexive, M_R has this form:

$$M_R = \begin{bmatrix} 0 & & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}$$ - main diagonal are "all 0"

➢ In digraph, R reflexive gives:



- All vertices have a loop. R irreflexive gives:



- all vertices have no loops.

Eg 34: Let A = {1, 2, 3, 4}

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M_T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

3) Symmetric

- If aRb, then bRa.

- matrix M_R is symmetric about diagonal.
- Digraph has no 1 way path, all paths are 2 ways.

4) Asymmetric

- If aRb, then b$\not R$a.

- matrix M_R is not symmetric & diagonal contains all 0's.
- digraph has no 2-way path.
- All paths are 1-way, no cycle of length 1.

5) Antisymmetric

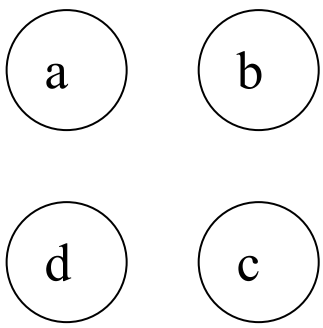
- If aRb & bRa, then a = b.
  If a $\neq$ b, then a$\not R$b or b$\not R$a.(contrapositive)

- loops allowed but must be 1-way path.

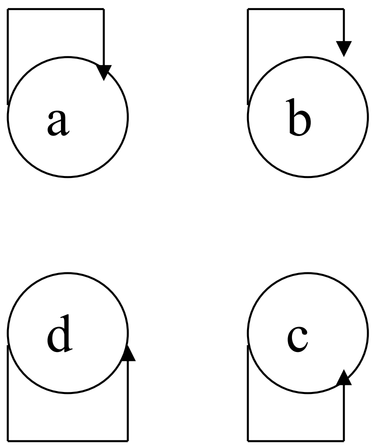
Eg 35: Let A = {1, 2, 3, 4}
R = {(1,2), (2,2), (3,4), (4,1)}

Eg 36: Let A = Z^+, aRb iff $a \mid b$.

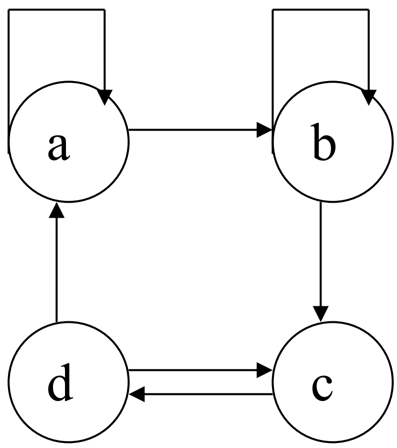
Eg 37: a)



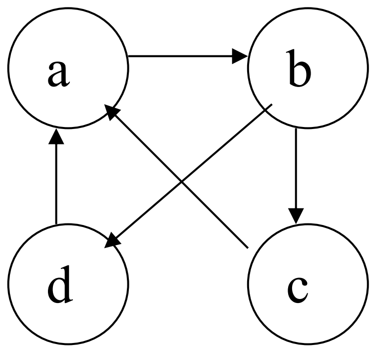
b)



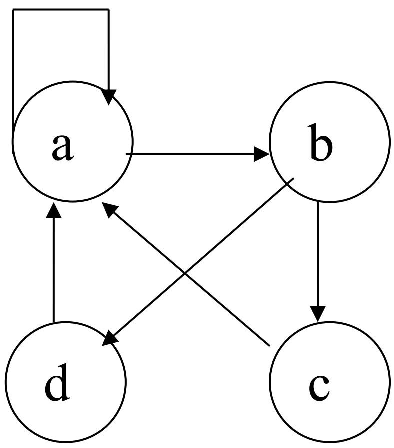
c)



d)



e)



Eg 38: a) Let A = {a, b, c}

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

b) Let A = {1, 2, 3, 4}

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

6) Transitive

- A relation R on set A is transitive aRb and bRc, then aRc.

Eg 39: Let A = Z, aRb iff a < b.

Eg 40: Let A = {1, 2, 3, 4},
R = {(1,2), (1,3), (4,2)}

Eg 41: $M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

Eg 42: Let A = {1, 2, 3},
R = {(1,1), (1,2), (1,3), (2,3), (3,3)}

Equivalence Relation

- R is an equivalence relation if it is
  - i) Reflexive
  - ii) Symmetric and
  - iii) Transitive

Eg 43: Let A be the set of all triangles in the plane and let R be the relation on A defined as follows:

$$R = \{(a,b) \in A \times A \mid a \text{ is congruent to } b\}$$

$\therefore$ R is an equivalence relation.

Eg 44: Let A = {1, 2, 3, 4} and let
R = {(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)}

Eg 45: Let A = Z, the set of integers, and let R be defined by aRb iff $a \leq b$. Is R an equivalence relation?

Eg 46: Let A = Z and let $R = \{(a,b) \in A \times A \mid a \equiv b \pmod{2}\}$.
Is R an equivalence relation?

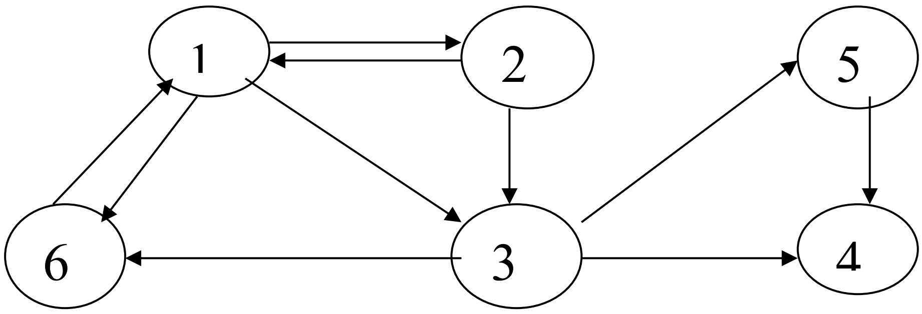
- If R is an equivalence relation on A, then the sets R(a) are called equivalence classes of R.
- Quotient set of A denoted by A/R is the partition consisting of all equivalence classes of R.

Eg 47: Let A = {1, 2, 3, 4}
R = {(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)}
Determine A/R.

Eg 48: Let A = {1, 2, 3, 4, 5, 6, 7}, then {{1, 3, 4, 5}, {2}, {6, 7}, is a partition of A. If two elements of A are related (equivalent) iff they are in the same subset, then a partition of the set A defines an equivalence relation on A.

Computer Representations

Eg 49:



Vertex	Tail	Head	Next

Eg 50: The following arrays describe a relation R on the set A = {1, 2, 3, 4, 5}. Compute both the digraph of R and matrix M_R.

VERT = [6, 2, 8, 7, 10]
TAIL = [2, 2, 2, 2, 1, 1, 4, 3, 4, 5]
HEAD= [4, 3, 5, 1, 2, 3, 5, 4, 2, 4]
NEXT =[3, 1, 4, 0, 0, 5, 9, 0, 0, 0]

Operations on Relations

- Let R and S be relations from a set A to a set B.

- R and S are subsets of A×B.

- Complement relation: $a\bar{R}b$ iff $a \not R b$.

- Inverse relation: $bR^{-1}a$ iff aRb.

- Intersection relation: $a(R \cap S)b$

- Union relation: $a(R \cup S)b$

Eg 51: Let A = {1, 2, 3, 4}, B = {a, b, c}

R = {(1,a), (1,b), (2,b), (2,c), (3,b), (4,a)}

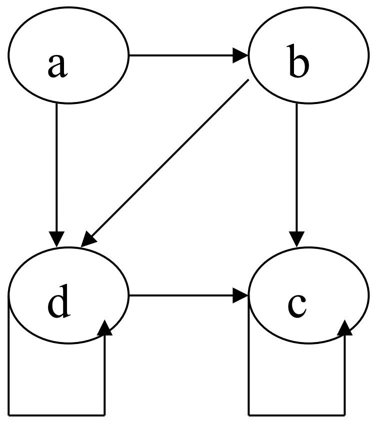
S = {(1,b), (2,c), (3,b), (4,b)}

Compute a) $\bar{R}$ b) $R \cap S$

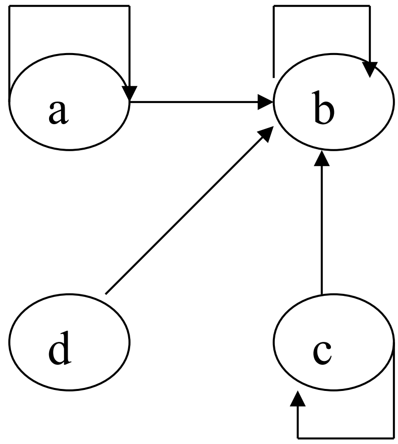
c) $R \cup S$ d) R^{-1}

Eg 52: Let A = {a, b, c, d} and R and S be two relations on A whose corresponding digraphs are as shown.

R



S



Compute $\overline{R}$, R^{-1} and $R \cap S$.

- $M_{R \cup S} = M_R \vee M_S$
- $M_{R \cap S} = M_R \wedge M_S$
- $M_{R^{-1}} = (M_R)^T$
- $(M_R)_{ij} = 1 \Leftrightarrow (M_{\overline{R}})_{ij} = 0$

Eg 53: Let A = {1, 2, 3} and let R and S be relations on A. Suppose that the matrices of R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Find $M_{\overline{R}}, M_{R^{-1}}, M_{R \cap S}$ and $M_{R \cup S}$.

TM: Suppose R and S are relations from A to B.

1) If $R \subseteq S$, then $R^{-1} \subseteq S^{-1}$ and $\overline{S} \subseteq \overline{R}$.
2) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$, $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$
3) $\overline{R \cap S} = \overline{R} \cup \overline{S}, \overline{R \cup S} = \overline{R} \cap \overline{S}$

TM: Let R and S be relations on a set A.

a) If R is reflexive, so is R^{-1}.
b) If R and S are reflexive, then so are $R \cap S$ and $R \cup S$.
c) R is reflexive if and only if $\bar{R}$ is irreflexive.

Eg 54: Let A ={1, 2, 3} and consider the two reflexive relations.

R = {(1,1), (1,2), (1,3), (2,2), (3,3)} and
S = {(1,1), (1,2), (2,2), (3,2), (3,3)}
Find R^{-1}, $\bar{R}$ and $R \cap S$.

TM: Let R be a relation on a set A. Then

a) R is symmetric iff $R = R^{-1}$.
b) R is antisymmetric iff $R \cap R^{-1} \subseteq \Delta$.
c) R is asymmetric iff $R \cap R^{-1} = \varnothing$.

TM: Let R and S be relations on A.

a) If R is symmetric, so are R^{-1} and $\bar{R}$.
b) If R and S are symmetric, so are $R \cap S$ and $R \cup S$.

Eg 55: Let $A = \{1, 2, 3\}$ and consider the symmetric relations
$R = \{(1,1), (1,2), (2,1), (1,3), (3,1)\}$
$S = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$

a) $R^{-1} =$

$\bar{R} =$

b) $R \cap S =$
$R \cup S =$

TM: Let R and S be relations on A.
a) $(R \cap S)^2 \subseteq R^2 \cap S^2$
b) If R and S are transitive, so is $R \cap S$.
c) If R and S are equivalence relations, so is $R \cap S$.

Composition relations

- The composition of R and S, written $S \circ R$.
  - If a is in A and c is in C,then $a(S \circ R)c$ iff for some b in B, we have aRb and bSc.

Eg 56: Let $A = \{1, 2, 3, 4\}$,
$R = \{(1,2), (1,1), (1,3), (2,4), (3,2)\}$ and
$S = \{ (1,4), (1,3), (2,3), (3,1), (4,1)\}$
Compute $S \circ R$.

Eg 57: Let A ={a, b, c} and let R and S be relations on A whose matrices are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix},\ M_S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Find S∘R.

Eg 58: Let A = {1, 2, 3, 4}, $M_R = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and

$M_S = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$. Find S∘R.

TM: Let A, B, C and D be sets, R a relation from A to B, S a relation from B to C, and T a relation from C to D. Then

$$T\circ (S\circ R) = (T\circ S) \circ R$$

TM: Let A, B and C be sets, R a relation from A to B, and Sa relation from B to C. Then

$$(S\circ R)^{-1} = R^{-1}\circ S^{-1}.$$

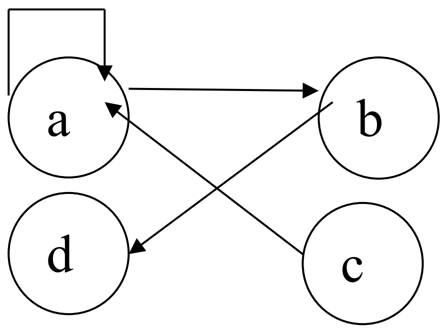
Closures

- Smallest relation R, that contains R and possesses the property we desire.

1) Reflexive closure of R = $R\cup\Delta$

Eg 59: A = {1, 2}, R = {(1,1), (1,2)}.
Reflexive closure of R =

Eg 60:

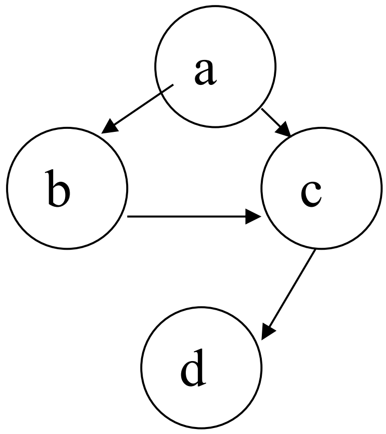


Not reflexive

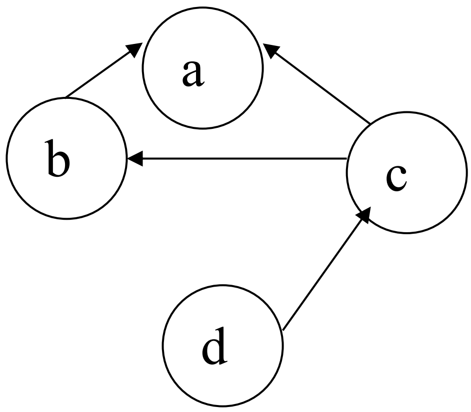
2) Symmetric closure
- The smallest symmetric relation containing R.
  Symmetric closure of R = $R \cup R^{-1}$

Eg 61: If A = {a, b, c, d},
R = {(a,b), (b,c), (a,c), (c,d)}
R^{-1} =
Symmetric closure of R =

Eg 62: R



R^{-1}



$R \cup R^{-1}$=

Eg 63: Matrix of symmetric closure,

$$M_{R \cup R^{-1}} = M_R \vee M_{R^{-1}}$$

3) Transitive closure
- The smallest transitive relation containing R.
- Let R be a relation on set A. Then R^{∞} is the transitive closure of R.

Eg 64: Let A = {1, 2, 3, 4}
R = {(1,2), (2,3), (3,4), (2,1)}
Find the transitive closure of R.

<u>Warshall Algorithm</u> (to find R^∞)

Let $w_0 = R$
Step 1: First transfer to w_k all 1's in w_{k-1}.
Step 2: List the locations $p_1, p_2, \ldots$ in column k of w_{k-1}, where the entry is 1,and the location $q_1, q_2, q_3, \ldots$ in row k w_{k-1}, where the entry is 1.
Step 3: Put 1's in all the position p_i, q_i of w_k (if they are not already there).

- Using this recurrence relation, find w_k for k=1, 2, 3,…,n.
- Then $w_n = R^\infty$ = transitive closure of R.

* If $|A| = 4$, we need not to consider more than M_{R^4} .

Eg 65: Apply Warshall Algorithm to

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ find } M_{R^\infty}.$$

- To find the transitive closure of R, the digraph method is impractical for large sets and relations and is not systematic.

- The matrix method is systematic enough to program for computer, but is inefficient and, for large matrices can be very costly.

- Fortunately, a more efficient algorithm, called Warshall's algorithm is available.

Eg 66: Let A = {1, 2, 3, 4, 5}
R = {(1,1), (1,2), (2,1), (2,2), (3,3), (3,4), (4,3), (4,4), (5,5)}
S = {(1,1), (2,2), (3,3), (4,4), (4,5), (5,4), (5,5)}
R and S are equivalence relation.
Find the smallest equivalence relation containing R and S by Warshall's algorithm.

TM: If R and S be two equivalence relations on a set A, then the smallest equivalence relation containing both R and S is $(R \cup S)^{\infty}$.
[equivalence closure of $R \cup S$]

Eg 67: Let A = {1, 2, 3, 4, 5} and let R and S be the equivalence relations on A whose matrices are given.
Compute the matrix of the smallest equivalence relation containing R and S and list the elements of this relation.

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \qquad M_S = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Chapter 4 Functions

Functions

- Let A and B be nonempty sets.
  A function *f* from A to B, which is denoted *f* : A→B, is a relation from A to B such that for all a∈Dom(*f*), *f*(a),the *f*-relative set of a, contains just one element of B.

Eg 1: Let A = {1, 2, 3, 4} and B = {a, b, c, d} and let
f = {(1,a), (2,a), (3,d), (4,c)}
f(1) = *f*(2) = *f*(3) = *f*(4) =

Eg 2: Let A = R be the set of real numbers and let
$p(x) = a_0 + a_1 x + \ldots + a_n x^n$ be a real polynomial.
⇒ The relation *p* is a function.

Eg 3: Let A = B = Z and let *f* : A→B be defined by
f(a) = a + 1, for a∈A.

Eg 4: Let A = Z and let B = {0, 1}. Let *f* : A→B be found by $f(a) = \begin{cases} 0 \; if \; a \; is \; even \\ 1 \; if \; a \; is \; odd \end{cases}$

- Suppose that *f* : A→B and *g* : B→C are functions. Then the composition of *f* and *g*, *g*∘*f* is a relation. *g*∘*f* also a function.

Eg 5: Let A = B = Z and C be the set of even integers.
Let f : A→B and g : B→C be defined by

$$f(a) = a + 1$$
$$g(b) = 2b$$

Find $g \circ f$.

Special types of functions

- Let f be a function from A to B.
  1) f is everywhere defined if Dom(f) = A.
  2) f is onto if Ran(f) = B.
  3) f is one to one if f(a) = f(a'), then a = a'.

Eg 6: Let A = $\{a_1, a_2, a_3\}$, B =$\{b_1, b_2, b_3\}$, C =$\{c_1, c_2\}$ and D = $\{d_1, d_2, d_3, d_4\}$. Consider the following four functions, from A to B, A to D, B to C and D to B respectively.

a) $f_1 = \{(a_1,b_2), (a_2,b_3), (a_3,b_1)\}$
b) $f_2 = \{(a_1,d_2), (a_2,d_1), (a_3,d_4)\}$
c) $f_3 = \{(b_1,c_2), (b_2,c_2), (b_3,c_1)\}$
d) $f_4 = \{(d_1,b_1), (d_2,b_2), (d_3,b_1)\}$

Determine whether each function is one to one, onto and everywhere defined.

- If f : A→B is a one to one function, then f assign to each element a of Dom(f) an element b = f(a) of Ran(f). Every b in Ran(f) is matched, with one and only one element of Dom(f).
  ∴ f is called a bijection between Dom(f) and Ran(f).

- If f is also everywhere defined and onto, then f is called a one to one correspondence between A and B.

Invertible Functions

- A function f : A→B is said to be invertible, f^{-1} is also a function.

Eg 7: Let f = {(1,a), (2,a), (3,d), (4,c)}

Then f^{-1} =

∴

TM: Let f : A→B be a function.

a) Then f^{-1} is a function from B to A if and only if f is 1-1.

If f^{-1} is a function, then

b) the function f^{-1} is also 1-1.

c) f^{-1} is everywhere defined if and only if f is onto.

d) f^{-1} is onto if and only if f is everywhere defined.

Eg 8: Let R be the set of real numbers and let f: R→R be defined by $f(x) = x^2$. Is f invertible?

Functions for Computer Science

Eg 9: Let A be a subset of the universal set U = {u_1, u_2, u_3, …, u_n}. The characteristic function of A is defined as a function from U to {0, 1} by the following:

$$f_A(u_i) = \begin{cases} 1 & if\ u_i \epsilon A \\ 0 & if\ u_i \notin A \end{cases}$$

If A = {4, 7, 9} and U = {1, 2, 3, …, 10},
then $f_A(2) = 0$, $f_A(4) = 1$, $f_A(7) = 1$ and $f_A(12) = \nexists$
∴ f_A is everywhere defined and onto, but is not 1-1.

Eg 10: Let A be the set of nonnegative integers, B = Z^+, and let f : A→B be defined by $f(n) = n!$

Eg 11: The floor function, which is defined for rational numbers as $f(q)$ is the largest integer less than or equal to q.

$f(1.5) = \lfloor 1.5 \rfloor$ $\quad$ $f(-3) = \lfloor -3 \rfloor$ $\quad$ $f(-2.7) = \lfloor -2.7 \rfloor$

= $\quad$ = $\quad$ =

Eg 12: The ceiling function, which is defined for rational numbers as c(q) is the smallest integer greater than or equal to q.

c(1.5) = $\lceil 1.5 \rceil$ c(-3) = $\lceil -3 \rceil$ c(-2.7) = $\lceil -2.7 \rceil$

= = =

Eg 13: a) Any polynomial with integer coefficients, *p*, can be used to define a function on Z as follows:
If $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$ and z∈Z, then *p*(z) is the value of *p* evaluated at z.

b) Let A = B = R and let f_n: A→B be defined for each positive integer n >1 as $f_n(x) = \log_n x$, the logarithm to the base n of *x*. In computer science applications, the bases 2 and 10 are particularly useful.

Permutation Functions

- A bijection from a set A to itself is called a permutation of A.

Eg 14: Let A = R and let *f* : A→A be defined by *f*(a) = 2a + 1.Since *f* is one-to-one and onto, it follows that *f* is a permutation of A.

- If A = {a_1, a_2, …, a_n} is a finite set and p is bijection on A.

$$\therefore\ p = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ p(a_1) & p(a_2) & \dots & p(a_n) \end{pmatrix}$$

Eg 15: Let A ={1, 2, 3}, then all the permutations of A are

$$I_A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix},\ P_1 = \begin{pmatrix} 1 & 2 & 3 \\ & & \end{pmatrix},\ P_2 = \begin{pmatrix} 1 & 2 & 3 \\ & & \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ & & \end{pmatrix},\ P_4 = \begin{pmatrix} 1 & 2 & 3 \\ & & \end{pmatrix},\ P_5 = \begin{pmatrix} 1 & 2 & 3 \\ & & \end{pmatrix}$$

Eg 16: Using the permutation of previous eg, compute

a) P_4^{-1} b) $P_3 \circ P_2$

Eg 17: Let A = {1, 2, 3, 4, 5}. The cycle (1, 3, 5) denotes

the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ & & & & \end{pmatrix}$

Eg 18: Let A = {1, 2, 3, 4, 5, 6}. Compute
(4, 1, 3, 5) o (5, 6, 3) and (5, 6, 3) o (4, 1, 3, 5).

- Two cycles of a set A are said to be disjoint if no element of A appears in both cycles.

Eg 19: Let A = {1, 2, 3, 4, 5, 6}, then the cycle (1, 2, 5) and (3, 4, 6) are _________ whereas the cycle (1, 2, 5) and (2, 4, 6) are ____________.

TM: A permutation of a finite set that is not the identity or a cycle can be written as a product of disjoint cycles of length ≥ 2.

Eg 20: Write the permutation

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{pmatrix}$$

of the set A = {1, 2, 3, 4, 5, 6, 7, 8} as a product of disjoint cycles.

- A cycle of length 2 is called a transposition.

- Every cycle can be written as a product of transpositions. In fact,

$$(b_1,b_2,\ldots,b_r) = (b_1,b_r) \circ (b_1,b_{r-1}) \circ\ldots\circ (b_1,b_3) \circ (b_1,b_2)$$

- Every permutation of a finite set with at least two elements can be written as a product of transpositions.

Eg 21: Write the permutation $p = (1, 2, 6) \circ (2, 4, 5) \circ (7, 8)$ as a product of transpositions.

TM: The product of transposition is not unique.

Even permutations

- Only as a product of an even number of transpositions.

Eg 22: (1, 3, 4, 2, 5) =

(3, 4, 2, 5, 1) =

containing odd number of elements in the cycle.

Odd permutations

- Only as a product of an odd number of transpositions.

Eg 23: (1, 3, 4, 2) =

(3, 4, 2, 1) =

containing even number elements in the cycle.

Eg 24: Is the permutations even or odd?

a) $p_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{pmatrix}$

b) $p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 4 & 5 & 7 & 6 & 3 & 1 \end{pmatrix}$

Chapter 5 Graph Theory

Graphs

A **graph** G consists of a finite set V of objects called **vertices**, a finite set E of objects called **edges**, and a function γ that assigns to each edge a subset $\{v, w\}$, where v and w are vertices (and may be the same).

We will write $G = (V, E, \gamma)$ when we need to name the parts of G. If e is an edge, and $\gamma(e) = \{v, w\}$, we say that e is an edge between v and w and that e is determined by v and w. The vertices v and w are called the end points of e. If there is only one edge between v and w, we often identify e with the set $\{v, w\}$. This should cause no confusion.

Example 1

Let $V = \{1,2,3,4\}$ and $E = \{e_1, e_2, e_3, e_4, e_5\}$.

Let γ be defined by

$\gamma(e_1) = \gamma(e_5) = \{1,2\}$, $\gamma(e_2) = \{4,3\}$,

$\gamma(e_3) = \{1,3\}$, $\gamma(e_4) = \{2,4\}$

Then $G = (V, E, \gamma)$ is a graph.

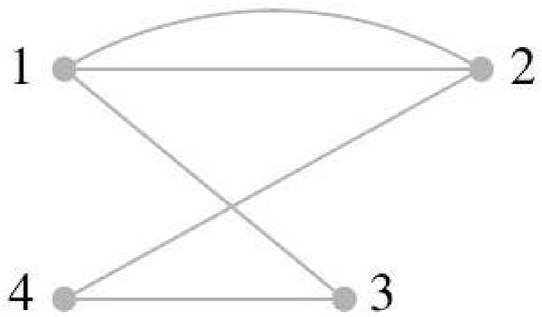


Figure 1

Example 2

Figures 2 and 3 also represent the graph G given in Example 1.

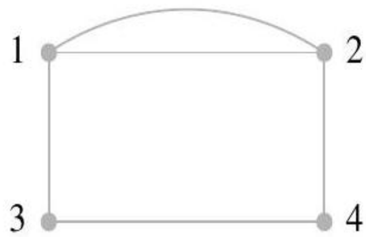


Figure 2

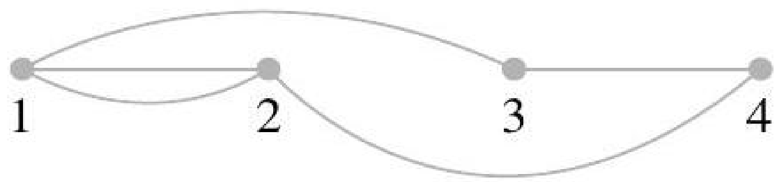


Figure 3

The **degree** of a vertex is the number of edges having that vertex as an end point.

A graph may contain an edge from a vertex to itself; such an edge is referred to as a loop. A loop contributes 2 to the degree of a vertex, since that vertex serves as both end points of the loop.

Example 3

(a) In the graph in Figure 4,

Vertex	A	B	C	D	E
Degree	2	4	1	3	2

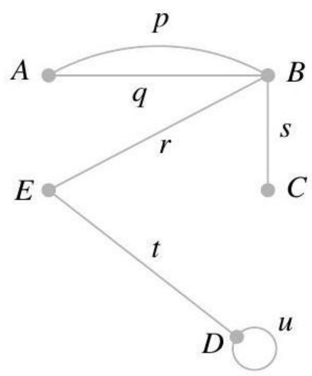


Figure 4

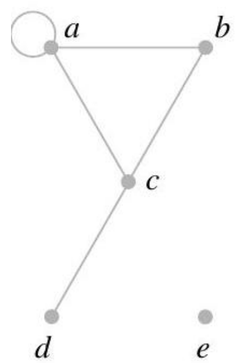


Figure 5

(b) In Figure 5, vertex a has degree 4, vertex e has degree 0, and vertex b has degree 2.

Vertex	a	b	c	d	e
Degree	4	2	3	1	0

(c) Each vertex of the graph in Figure 6 has degree 2.

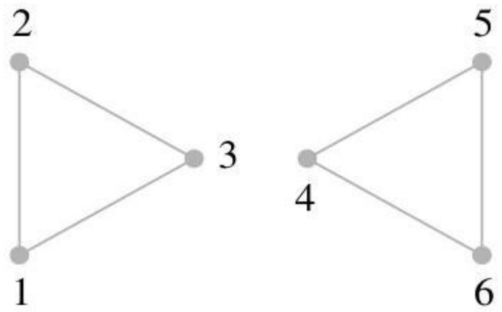


Figure 6

A vertex with degree 0 is called an **isolated** vertex. A pair of vertices that determine an edge are **adjacent** vertices.

<u>Example 4</u>
In Figure 5, vertex e is an isolated vertex. In Figure 5, a and b are adjacent vertices; vertices a and d are not adjacent.

A **path** π in a graph $\boldsymbol{G}$ consists of a pair (V_π, E_π) of sequences: a vertex sequence $V_\pi: v_1, v_2, \dots, v_k$ and an edge sequence $E_\pi: e_1, e_2, \dots, e_{k-1}$ for which

1. Each successive pair v_i, v_{i+1} of vertices is adjacent in G, and edge e_i has v_i and v_{i+1} as end points, for $i = 1, \dots, k-1$.
2. No edge occurs more than once in the edge sequence.

Thus we may begin at v_1 and travel through the edges $e_1, e_2, \ldots, e_{k-1}$ to v_k without using any edge twice.

A **circuit** is a path that begins and ends at the same vertex.

Such paths may be called **cycles**; the word "circuit" is more common in general graph theory.

A path is called **simple** if no vertex appears more than once in the vertex sequence, except possibly if $v_1 = v_k$. In this case the path is called a simple circuit.

<u>Example 5</u>

(a) For the graph in Figure 4 we define

a path π_1 by sequences $V_{\pi_1}: A, B, E, D, D$ and $E_{\pi_1}: p, r, t, u$, and

the path π_2 by the sequences $V_{\pi_2}: A, B, A$ and $E_{\pi_2}: p, q$.

Path π_1 is not simple, since vertex D appears twice, but π_2 is a simple circuit, since the only vertex appearing twice occurs at the beginning and at the end.

(b) The path $\pi_4: D, E, B, C$ in the graph of Figure 4 is simple. Here no mention of edges is needed.

(c) Examples of paths in the graph of Figure 5 are $\pi_5: a, b, c, a$ and $\pi_6: d, c, a, a$. Here π_5 is a simple circuit, but π_6 is not simple.

(d) In Figure 6 the vertex sequence 1, 2, 3, 2 does not specify a path, since the single edge between 2 and 3 would be traveled twice.

(e) The path π_7: c, a, b, c, d in Figure 5 is not simple.

Euler Paths and Circuits

A path in a graph G is called an **Euler path** if it includes every edge exactly once.

An **Euler circuit** is an Euler path that is a circuit.

Leonhard Euler (1707-1783) worked in many areas of mathematics. The names "Euler path" and "Euler circuit" recognize his work with the Königsberg Bridge problem.

Example 6

(a) An Euler path in Figure 26 is π: E, D, B, A, C, D. A little experimentation will show that no Euler circuit is possible for this graph. (Why?)

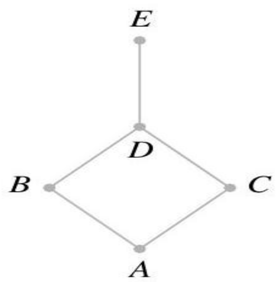


Figure 26

(b) One Euler circuit in the graph of Figure 27 is π : 5, 3, 2, 1, 3, 4, 5 .

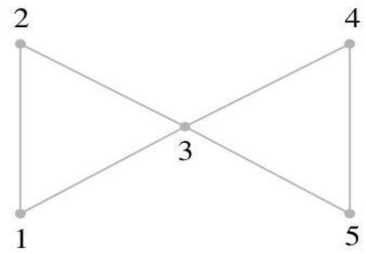


Figure 27

Two questions arise naturally at this point. Is it possible to determine whether an Euler path or Euler circuit exists without actually finding it? If there must be an Euler path or circuit, is there an efficient way to find one?

Consider again the graphs in Example 6. In Figure 26 the edge $\{D, E\}$ must be either the first or the last traveled, because there is no other way to travel to or from vertex E. This means that if G has a vertex of degree 1, there cannot be an Euler circuit, and if there is an Euler path, it must begin or end at this vertex. A similar argument applies to any vertex v of odd degree, say $2n + 1$. We may travel in on one of these edges and out on another one n times, leaving one edge from v untraveled. This last edge may be used for leaving v or arriving at v, but not both, so a circuit cannot be completed. We have just shown the first of the following results.

<u>Theorem 8.1</u>
(a) If a graph G has a vertex of odd degree, there can be no Euler circuit in G.

(b) If G is a connected graph and every vertex has even degree, then there is an Euler circuit in G.

We have proved that if G has vertices of odd degree, it is not possible to construct an Euler circuit for G, but an Euler path may be possible. Our earlier discussion noted that a vertex of odd degree must be either the beginning or the end of any possible Euler path. We have the following theorem.

Theorem 8.2

(a) If a graph G has more than two vertices of odd degree, then there can be no Euler path in G.

(b) If G is connected and has exactly two vertices of odd degree, there is an Euler path in G. Any Euler path in G must begin at one vertex of odd degree and end at the other.

Example 7

Which of the graphs in Figures 30, 31, and 32 have an Euler circuit, an Euler path but not an Euler circuit, or neither?

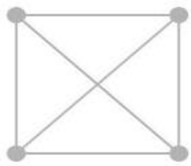


Figure 30

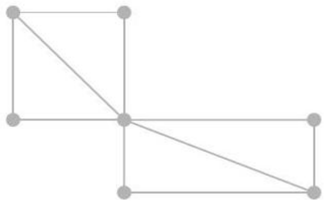


Figure 31

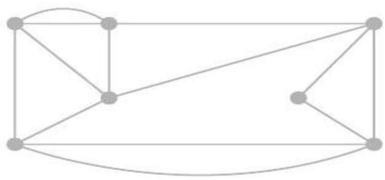


Figure 32

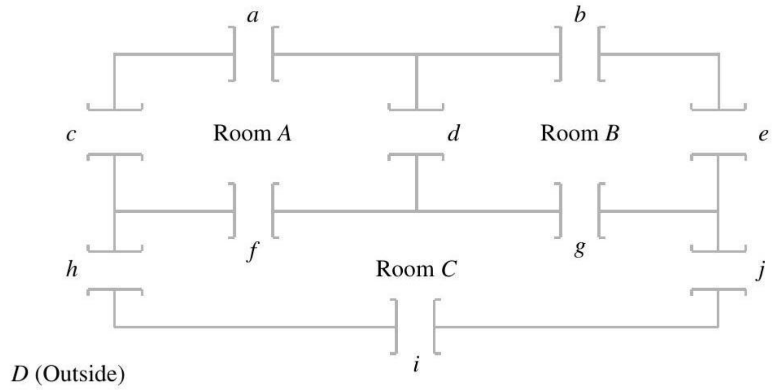
Solution

(a) In Figure 30, each of the four vertices has degree 3; thus, by Theorems 1 and 2, there is neither an Euler circuit nor an Euler path.

(b) The graph in Figure 31 has exactly two vertices of odd degree. There is no Euler circuit, but there must be an Euler path.

(c) In Figure 32, every vertex has even degree; thus the graph must have an Euler circuit

Example 8



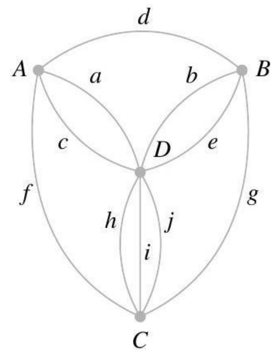


Figure 28 Figure 29

Consider the floor plan of a three-room structure that is shown in Figure 28. Each room is connected to every room that it shares a wall with and to the outside along each wall. The problem is this: Is it possible to begin in a room or outside and take a walk that goes through each door exactly once? This diagram can also be formulated as a graph where each room and the outside constitute a vertex and an edge corresponds to each door. A possible graph for this structure is shown in Figure 29. The translation of the problem is whether or not there exists an Euler path for this graph.

Vertex	A	B	C	D
Degree	4	4	5	5

This graph has exactly 2 vertices of odd degree. Thus the problem can be solved by Theorem 8.2; that is, there is an Euler path. One is shown in Figure 33. Using the labels of Figure 29, this path π is specified by

$V_\pi: C, D, C, A, D, A, B, D, B, C, D$ and $E_\pi: i, h, f, c, a, d, b, e, g, j$.

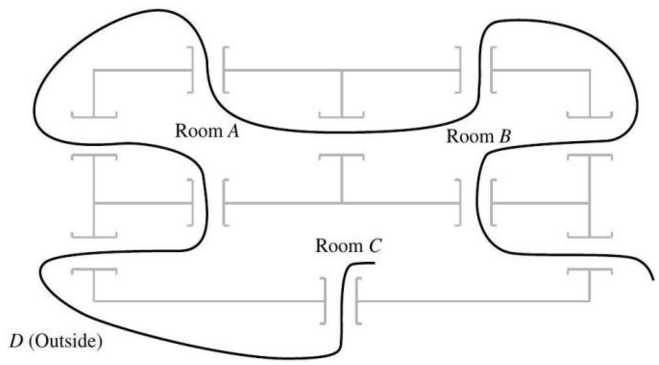


Figure 33

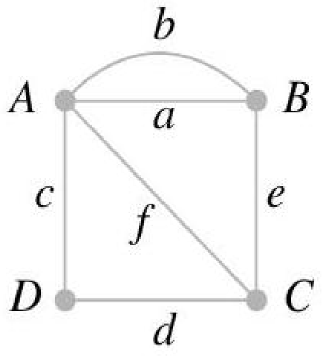


Figure 34

Vertex	A	B	C	D
Degree	4	3	3	2

In Figure 34, an Euler path must begin (or end) at B and end (or begin) at C. One possible Euler path for this graph is specified by $V_\pi: B, A, D, C, A, B, C$ and $E_\pi: a, c, d, f, b, e$.

Hamiltonian Paths and Circuits

A **Hamiltonian path** is a path that contains each vertex exactly once.

A **Hamiltonian circuit** is a circuit that contains each vertex exactly once except for the first vertex, which is also the last.

Example 9

Consider the graph in Figure 55. The path a, b, c, d, e is a Hamiltonian path because it contains each vertex exactly once. It is not hard to see, however, that there is no Hamiltonian circuit for this graph. For the graph shown in Figure 56, the path A, D, C, B, A is a Hamiltonian circuit. In Figures 57 and 58, no Hamiltonian path is possible. (Verify this.)

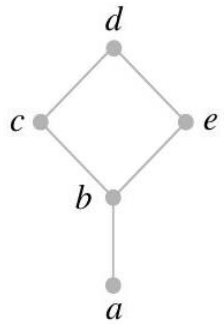


Figure 55

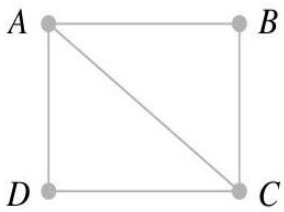


Figure 56

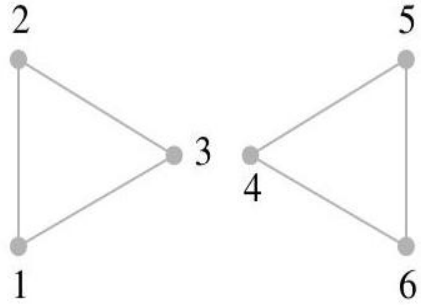


Figure 57

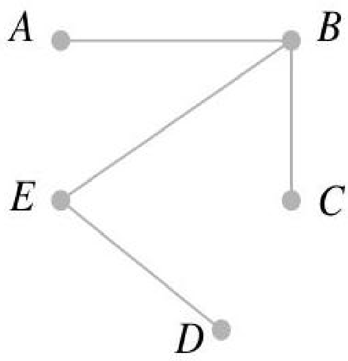


Figure 58

Questions analogous to those about Euler paths and circuits can be asked about Hamiltonian paths and circuits. Is it possible to determine whether a Hamiltonian path or circuit exists? If there must be a Hamiltonian path or circuit, is there an efficient way to find it? Surprisingly, considering Theorems 8.1 and 8.2, the first question about Hamiltonian paths and circuits has not been completely answered and the second is still unanswered as well. However, we can make several observations based on the examples.

If a graph G on n vertices have a Hamiltonian circuit, then G must have at least n edges.

We now state some partial answers that say if a graph G has "enough" edges, a Hamiltonian circuit can be found. These are again existence statements; no method for constructing a Hamiltonian circuit is given.

Theorem 8.3

Let G be a connected graph with n vertices, $n > 2$, and no loops or multiple edges. G has a Hamiltonian circuit if for any two vertices u and v of G that are not adjacent, the degree of u plus the degree of v is greater than or equal to n.

Corollary 8.4

G has a Hamiltonian circuit if each vertex has degree greater than or equal to $n/2$.

Chapter 6 Group Theory

Binary Operations

A **binary operation** on a set A is an everywhere defined function $f: A \times A \rightarrow A$.

Observe the following properties that a binary operation must satisfy:

1. Since Dom $(f) = A \times A, f$ assigns an element $f(a, b)$ of A to each ordered pair (a, b) in $A \times A$. That is, the binary operation must be defined for each ordered pair of elements of A.
2. Since a binary operation is a function, only one element of A is assigned to each ordered pair.

<u>Example 1</u>

Let $A = \mathbb{Z}$. Define $a * b$ as $a + b$. Then $*$ is a binary operation on $\mathbb{Z}$.

<u>Example 2</u>

Let $A = \mathbb{R}$. Define $a * b$ as a/b. Then $*$ is not a binary operation, since it is not defined for every ordered pair of elements of A. For example, $3 * 0$ is not defined, since we cannot divide by zero, that is 3/0 = undefined.

Example 3

Let $A = \mathbb{Z}^{+}$. Define $a * b$ as $a - b$. Then $*$ is not a binary operation since it does not assign an element of A to every ordered pair of elements of A; for example, $2 * 5 = 2 - 5 = -3 \notin A$.

Group

A **group** is a set together with a binary operation $*$ on G such that

[Associative]

1. $(a * b) * c = a * (b * c)$ for any elements a, b, and c in G.

[Commutative]

2. There is a unique element e in G such that

$$a * e = e * a \quad \text{for any } a \in G$$

Note: element e is called an identity element.

[Inverse]

3. For every $a \in G$, there is an element $a' \in G$, called an inverse of a, such that

$$a * a' = a' * a = e$$

Observe that if $(G,*)$ is a group, then $*$ is a binary operation, so G must be closed under $*$; that is,

$$a * b \in G \text{ for any elements } a \text{ and } b \text{ in } G$$

To simplify our notation, from now on when only one group $(G,*)$ is under consideration and there is no possibility of confusion, we shall write the product $a * b$ of the elements a and b in the group $(G,*)$ simply as ab, and we shall also refer to $(G,*)$ simply as G.

Abelian

A group G is said to be **Abelian** if $ab = ba$ for all elements a and b in G.

<u>Example 4</u>
The set of all integers $\mathbb{Z}$ with the operation of ordinary addition is an Abelian group. If $a \in \mathbb{Z}$, then an inverse of a is its opposite $-a$.

<u>Example 5</u>
The set $\mathbb{Z}^+$ under the operation of ordinary multiplication is not a group since, for example, the element 2 in $\mathbb{Z}^+$ has no inverse.

Example 6

The set of all nonzero real numbers under the operation of ordinary multiplication is a group. An inverse of $a \neq 0$ is $1/a$.

Example 7

Let G be the set of all nonzero real numbers and let

$$a * b = \frac{ab}{2}$$

Show that $(G,*)$ is an Abelian group.

Solution

We first verify that $*$ is a binary operation. If a and b are elements of G, then $\frac{ab}{2}$ is a nonzero real number and hence is in G. We next verify associativity. Since

$$(a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{(ab)c}{4}$$

and since

$$a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{a(bc)}{4} = \frac{(ab)c}{4}$$

the operation $*$ is associative.

The number 2 is the identity in G, for if $a \in G$, then

$$a * 2 = \frac{(a)(2)}{2} = a = \frac{(2)(a)}{2} = 2 * a$$

the identity element, $e = 2$.

Finally, if $a \in G$, then $a' = 4/a$ is an inverse of a, since

$$a * a' = a * \frac{4}{a} = \frac{a(4/a)}{2} = 2 = \frac{(4/a)(a)}{2} = \frac{4}{a} * a = a' * a.$$

Since $a * b = b * a$ for all a and b in G, we conclude that G is an Abelian group.

Subgroup

Let H be a subset of a group G such that

(a) The identity e of G belongs to H.

(b) If a and b belong to H, then $ab \in H$.

(c) If $a \in H$, then $a^{-1} \in H$.

Then H is called a **subgroup** of G.

Group Code

The basic unit of information, called a **message**, is a finite sequence of characters from a finite alphabet. We shall choose as our alphabet the set $B = \{0,1\}$.

The set B is a group under the binary operation + whose table is shown in below

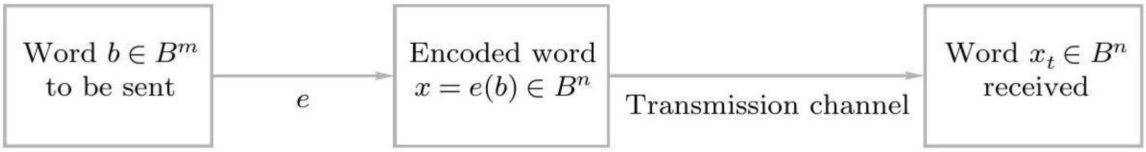
+	0	1
0	0	1
1	1	0

If we think of B as the group $\mathbb{Z}_2$, then + is merely mod 2 addition. It follows that $B^m = B \times B \times \cdots \times B$ (m factors) is a group under the operation $\oplus$ defined by

$$(x_1, x_2, \dots, x_m) \oplus (y_1, y_2, \dots, y_m) = (x_1 + y_1, x_2 + y_2, \dots, x_m + y_m)$$

This group's identity is $\overline{0} = (0,0,\dots,0)$ and every element is its own inverse. An element in B^m will be written as $(b_1, b_2, \dots, b_m)$ or more simply as $b_1 b_2 \cdots b_m$.

Observe that B^m has 2^m elements. That is, the order of the group B^m is 2^m.



If the transmission channel is noiseless, then $x_t = x$ for all x in B^n. In this case $x = e(b)$ is received for each $b \in B^m$, and since e is a known function, b may be identified.

In general, errors in transmission do occur. We will say that the code word $x = e(b)$ has been transmitted with $\boldsymbol{k}$ or fewer errors if x and x_t differ in at least 1 but no more than k positions.

Let $e: B^m \to B^n$ be an (m, n) encoding function. We say that e detects k or fewer errors if whenever $x = e(b)$ is transmitted with k or fewer errors, then x_t is not a code word (thus x_t could not be x and therefore could not have been correctly transmitted).

For $x \in B^n$, the number of 1's in x is called the **weight** of x and is denoted by $|x|$.

Example 8

Find the weight of each of the following words in B^5 :

(a) $x = 0\mathbf{1}000$

(b) $x = \mathbf{111}00$

(c) $x = 00000$

(d) $x = \mathbf{11111}$

Solution

(a) $|x| = 1$

(b) $|x| = 3$

(c) $|x| = 0$

(d) $|x| = 5$

Let x and y be words in B^m. The **Hamming distance** $\delta(x, y)$ between x and y is the weight, $|x \oplus y|$, of $x \oplus y$.

Thus the distance between $x = x_1x_2 \cdots x_m$ and $y = y_1y_2 \cdots y_m$ is the number of values of i such that $x_i \neq y_i$, that is, the number of positions in which x and y differ.

Using the weight of $x \oplus y$ is a convenient way to count the number of different positions.

Example 9

Find the distance between x and y :

(a) $x = 110110, y = 000101$

(b) $x = 001100, y = 010110$

Solution

(a) $x \oplus y = 110011$, so $|x \oplus y| = 4$

(b) $x \oplus y = 011010$, so $|x \oplus y| = 3$

The **minimum distance** of an encoding function $e: B^m \to B^n$ is the minimum of the distances between all distinct pairs of code words; that is,

$$\min\{\delta(e(x), e(y)) \mid x, y \in B^m\}$$

THEOREM 7.1

An (m, n) encoding function $e: B^m \to B^n$ can detect k or fewer errors if and only if its minimum distance is at least $k + 1$.

An (m, n) encoding function $e: B^m \to B^n$ is called a **group code** if

$$e(B^m) = \{e(b) \mid b \in B^m\} = \text{Ran } (e)$$

is a subgroup of B^n.

THEOREM 7.2

Let $e: B^m \to B^n$ be a group code. The minimum distance of e is the minimum weight of a nonzero code word.

Example 10

Let $m = 2$ and $n = 5$. Consider the encoding function $e : B^2 \rightarrow B^5$ defined by

$$e(00) = 00000$$

$$e(10) = 10110$$

$$e(01) = 01011$$

$$e(11) = 11101$$

(a) Verify that the (2, 5) encoding function e: $B^2 \rightarrow B^5$ given above is a group code.
(b) Then, find the minimum distance of this group code.
(c) How many errors will e detects?

Solution

a) To verify is a group code:

$\oplus$	00000	10110	01011	11101
00000	00000	10110	01011	11101
10110	10110	00000	11101	01011
01011	01011	11101	00000	10110
11101	11101	01011	10110	00000

Let $G = \{00000, 10110, 01011, 11101\}$.

00000 is the identity of B^5 and it is belong to G.

If x and y are elements of G, then $x \oplus y$ is in G.

[G is closed under operation $\oplus$]

Subgroup
Let H be a subset of a group G such that

(a) The identity e of G belongs to H.

(b) If a and b belong to H, then $ab \in H$.

(c) If $a \in H$, then $a^{-1} \in H$.

Then H is called a **subgroup** of G.

Each element is it's own inverse.

(00000' = 00000; 10110' = 10110; 01011' = 01011'; 11101' = 11101)

G is a subgroup of B^5. Therefore $(G,\oplus)$ is a group code.

b)

Codeword	10110	01011	11101
Weight	3	3	4

The minimum distance = minimum weight

= **3**.

c)

Thus the group code will detect (**3** - 1) = 2 or fewer errors.

Example 10(**Method 2**)

Let $m = 2$ and $n = 5$. Consider the encoding function e : $B^2 \to B^5$ defined by

$$e(00) = 00000$$

$$e(10) = 10110$$

$$e(01) = 01011$$

$$e(11) = 11101$$

(a) Verify that the (2, 5) encoding function e: $B^2 \to B^5$ given above is a group code.

(b) Then, find the minimum distance of this group code.

(c) How many errors will e detects?

Solution

(a) To verify is a group code:

$$\text{Let } e(00) = 00000 = C_0$$

$$e(10) = 10110 = C_1$$

$$e(01) = 01011 = C_2$$

$$e(11) = 11101 = C_3$$

$\oplus$	$C_0 = 00000$	$C_1 = 10110$	$C_2 = 01011$	$C_3 = 11101$
$C_0 = 00000$	C_0	C_1	C_2	C_3
$C_1 = 10110$	C_1	C_0	C_3	C_2
$C_2 = 01011$	C_2	C_3	C_0	C_1
$C_3 = 11101$	C_3	C_2	C_1	C_0

Let $G = \{C_0, C_1, C_2, C_3\}$.

C_0 = 00000 is the identity of B^5 and it is belong to G.

If C_i and C_j are elements of G, then $C_i \oplus C_j = C_k$ is in G.

[G is closed under operation ⊕]

Subgroup
Let H be a subset of a group G such that

(a) The identity e of G belongs to H.

(b) If a and b belong to H, then $ab \in H$.

(c) If $a \in H$, then $a^{-1} \in H$.

Then H is called a **subgroup** of G.

Each element is it's own inverse.

($C_0{}' = C_0$; $C_1{}' = C_1$; $C_2{}' = C_2$; $C_3{}' = C_3$)

G is a subgroup of B^5. Therefore $(G, \oplus)$ is a group code.

Find minimum distance from nonzero code word.

b)

Codeword	~~00000~~	10110	01011	11101
Weight	~~0~~	**3**	**3**	4

The minimum distance = minimum weight

= **3**.

c)

Thus the group code will detect (**3** - 1) = 2 or fewer errors.

Example 11(**Method 2**)

Consider the (3,6) encoding function $e : B^3 \to B^6$ defined by

e(000) = 000 000
e(001) = 001 100
e(010) = 010 011
e(100) = 100 101
e(101) = 101 001
e(110) = 110 110
e(111) = 111 010

(a) Show that this encoding function is a group code.

(b) Then, find the minimum distance of this group code.

(c) How many errors will *e* detects?

Solution

Let G = $\{c_0, c_1, c_2, c_3, c_4, c_5, c_6, c_7\}$

e(000) = 000 000 = c_0
e(001) = 001 100 = c_1
e(010) = 010 011 = c_2
e(011) = 011 111 = c_3
e(100) = 100 101 = c_4
e(101) = 101 001 = c_5
e(110) = 110 110 = c_6
e(111) = 111 010 = c_7

$\oplus$	c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7
c_0	c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7
c_1	c_1	c_0	c_3	c_2	c_5	c_4	c_7	c_6
c_2	c_2	c_3	c_0	c_1	c_6	c_7	c_4	c_5
c_3	c_3	c_2	c_1	c_0	c_7	c_6	c_5	c_4
c_4	c_4	c_5	c_6	c_7	c_0	c_1	c_2	c_3
c_5	c_5	c_4	c_7	c_6	c_1	c_0	c_3	c_2
c_6	c_6	c_7	c_4	c_5	c_2	c_3	c_0	c_1
c_7	c_7	c_6	c_5	c_4	c_3	c_2	c_1	c_0

C_0 = 000000 is the identity of B^6 and it is belong to G.

If C_i and C_j are elements of G, then $C_i \oplus C_j = C_k$ is in G.

[G is closed under operation $\oplus$]

Each element is it's own inverse.

($C_0' = C_0$; $C_1' = C_1$; $C_2' = C_2$; $C_3' = C_3$;

$C_4{}' = C_4$; $C_5{}' = C_5$; $C_6{}' = C_6$; $C_7{}' = C_7$)

G is a subgroup of B^6. Therefore $(G,\oplus)$ is a group code

b)

Codeword	~~000000~~	001 100	010 011	011 111
Weight	~~0~~	**2**	3	5
Codeword	100 101	101 001	110 110	111 010
Weight	3	3	4	4

The minimum distance = minimum weight

= **2**.

c)

Thus the group code will detect (**2** - 1) = 1 or fewer error.

Chapter 7 Boolean Algebra

- Boolean algebra is a branch of mathematics concerned with a logical calculus in which structures and rules are applied to logical symbols in the same way that ordinary algebra is applied to symbols representing numerical quantities.
- Boolean variables are the variables that can take only value 0 or 1.

<u>Properties of Boolean algebra</u>

1) $(x')' = x$ Involution
2) $(x \wedge y)' = x' \vee y'$ DeMorgan's law
3) $(x \vee y)' = x' \wedge y'$ DeMorgan's law

- A Boolean Algebra can also be considered as a set of mathematical objects with 2 operations + and *(or, $\vee$ and $\wedge$) defined on it satisfying the following axioms:

A1) Commutative laws

$x+y = y+x, \quad x*y = y*x$

A2) Associative laws

$(x+y)+z = x+(y+z)$

$(x*y)*z = x*(y*z)$

A3) Distributive laws

$x+(y*z) = (x+y)*(x+z)$

$x*(y+z) = (x*y)+(x*z)$

A4) Identity laws

There exist 0 and 1 such that $x + 0 = x$ and $x * 1 = x, \forall x$.

A5) Complement

For every x, there is an x' such that $x+x'=1$, $x*x'=0$. x' is called the complement of x.

Eg 6: Set of sets with operations of $\vee$ and $\wedge$ form a Boolean algebra with

$+ = \cup$ $\quad$ $1 = U$

$* = \cap$ $\quad$ $0 = \varnothing$

Eg 7: Set of Propositions with operations of $\vee$ and $\wedge$ form a Boolean algebra with

$+ = \vee$ $\quad$ $0 = F$

$* = \wedge$ $\quad$ $1 = T$

Eg 8: Boolean algebra B with two elements 0 and 1 (called bits). Let + and * be the operations defined as:

+	1	0
1	1	1
0	1	0

*	1	0
1	1	0
0	0	0

Eg 9: Simplify the following Boolean expressions.

a) $E(x,y,z) = (x+yz)(x'y'+z)$

b) $E(x,y,z) = (x+y'z')'$

Boolean Polynomial or Boolean expression

- Let $x_1, x_2, \ldots, x_n$ be a set of n symbols or variables. A Boolean polynomial $p(x_1, x_2, \ldots, x_n)$ is defined as follows:

1) $x_1, x_2, \ldots, x_n$ are all Boolean Polynomials.
2) The symbols 0 and 1 are Boolean polynomials.
3) The $p(x_1, x_2, \ldots, x_n)$ and $q(x_1, x_2, \ldots, x_n)$ are Boolean polynomials. Then are $p(x_1, x_2, \ldots, x_n)$ $\wedge$or$\vee$ $q(x_1, x_2, \ldots, x_n)$
4) If p is a Boolean polynomial, so is p'.
5) There are no Boolean polynomials in the variables x_k other than those that can be obtained by repeated use of rules (1), (2), (3) and (4).

Eg 10: $p(x,y,z) = (x \wedge y) \vee z$
$= (x * y) + z$

$p(x,y,z) = (x \vee y') \vee (y \wedge 1)$
$= (x + y') + (y * 1)$

Eg 11: $(x' \wedge y' \wedge z') \vee (x' \wedge y' \wedge z) \vee (x \wedge y' \wedge z') \vee (x \wedge y' \wedge z)$
$=$

A Boolean function is a function that takes values from B_n to produce a value in B.

$$f : B_n \rightarrow B$$

$\Rightarrow$ f is a function of n variables $x_1, x_2, \ldots, x_n$ each, may take value 0 or 1 and produces a value which is either 0 or 1.

$\Rightarrow$ f represents a function that produce an output of 0 or 1 from n inputs of 0 or 1.

Eg 12: Boolean function may be described by a truth table.

x_1	x_2	x_3	$f(x_1,x_2,x_3)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

The function $f(x_1, x_2, \ldots, x_n)$

=

TM: Any function $f : B_n \rightarrow B$ is produced by a Boolean expression.

Eg 13: Consider the function f : $B_3 \to B$ with truth table below

x	y	z	$f(x,y,z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

f is produced by the Boolean expression.
$E(x,y,z) =$

This Boolean expression can be simplified to
$E(x,y,z) =$

Eg 14: A function $f(x,y,z)$ has truth table below, write the **minterm expression** representing $f(x,y,z)$. Simplify the expression.

x	y	z	$f(x,y,z)$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Karnaugh Map

- A pictorial device for finding prime implicants and minimal forms for Boolean expressions.

Case of two variables

	y'	y
x'	$x'y'$	$x'y$
x	xy'	xy

		(a)	(b)	(c)	(d)	(e)
x	y	$f(x,y)$	$f(x,y)$	$f(x,y)$	$f(x,y)$	$f(x,y)$
0	0	1	1	0	1	1
0	1	1	0	0	1	0
1	0	0	1	1	1	0
1	1	0	0	1	0	1

a)

	y'	y
x'		
x		

b)

	y'	y
x'		
x		

c)

	y'	y
x'		
x		

d)

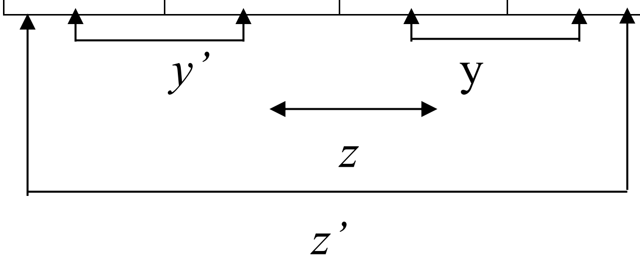
	y'	y
x'		
x		

e)

	y'	y
x'		
x		

Case of three variables

	$y'z'$	$y'z$	yz	yz'
x'				
x				



a)

	$y'z'$	$y'z$	yz	yz'
x'	1	1		
x	1	1		

b)

	$y'z'$	$y'z$	yz	yz'
x'			1	1
x			1	1

c)

	$y'z'$	$y'z$	yz	yz'
x'		1	1	
x		1	1	

d)

	$y'z'$	$y'z$	yz	yz'
x'	1			1
x	1			1

e)

	$y'z'$	$y'z$	yz	yz'
x'	1	1	1	1
x				

f)

	$y'z'$	$y'z$	yz	yz'
x'				
x	1	1	1	1

g)

	$y'z'$	$y'z$	yz	yz'
x'	1			
x	1			

h)

	$y'z'$	$y'z$	yz	yz'
x'		1	1	
x				

i)

	$y'z'$	$y'z$	yz	yz'
x'	1			1
x				

Eg 15:

a) $p(x,y,z) = x'y'z' + x'y'z + xy'z' + xy'z$

$= x'y'(z'+z) + xy'(z'+z)$

$= x'y' + xy'$

$= y'(x'+x)$

$= y'$

g) $p(x,y,z) =$

i) $p(x,y,z) =$

Eg 16:

			(a)	(b)	(c)
x	y	z	$f(x,y,z)$	$f(x,y,z)$	$f(x,y,z)$
0	0	0	1	1	1
0	0	1	1	0	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	0	1	0
1	1	1	1	1	1

a)

	$y'z'$	$y'z$	yz	yz'
x'				
x				

	$y'z'$	$y'z$	yz	yz'
x'				
x				

b)

	$y'z'$	$y'z$	yz	yz'
x'				
x				

c)

	$y'z'$	$y'z$	yz	yz'
x'				
x				

Eg 17: A function $f(x,y,z)$ has the truth table below:

x	y	z	$f(x,y,z)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

i) Write the minterm expression representing $f(x,y,z)$.
ii) Show that, by constructing Karnaugh map or otherwise, the expression can be simplified to $(x \vee z) \wedge y'$.

(i)

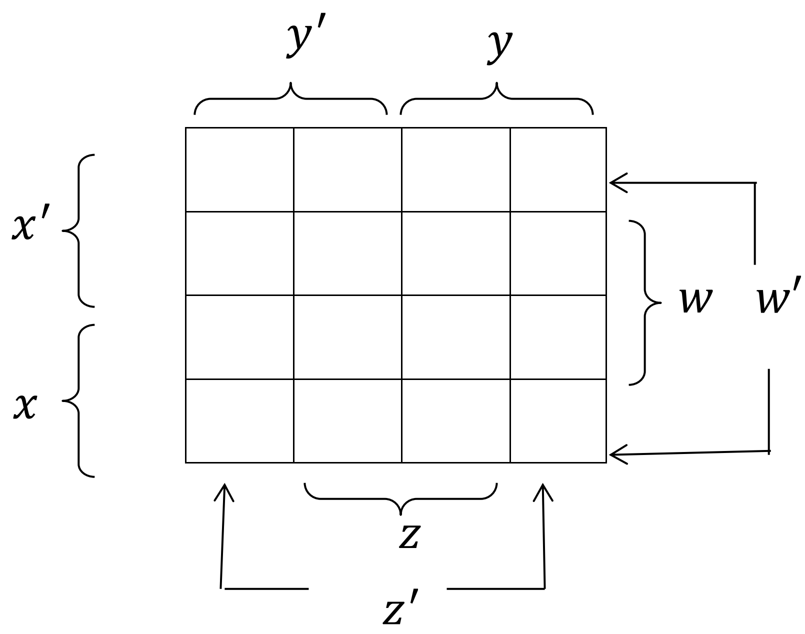
(ii)

	$y'z'$	$y'z$	yz	yz'
x'				
x				

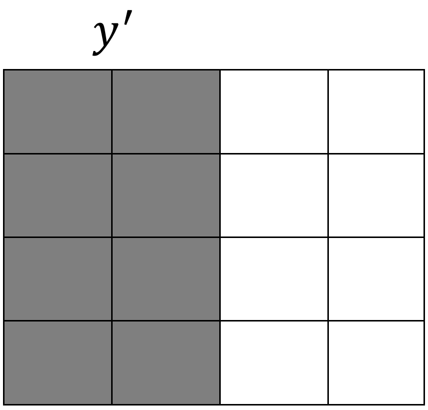
Case of four variables

f : B$_4$ $\rightarrow$ B, considered as a function of x, y, z and w.

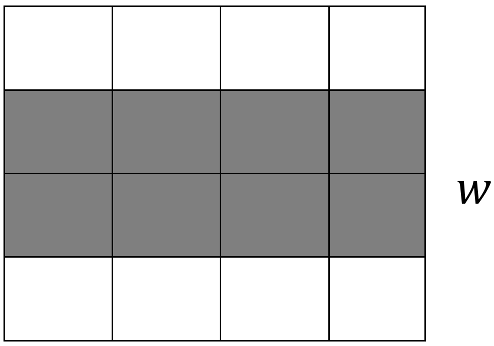
	$y'z'$	$y'z$	yz	yz'
$x'w'$				
$x'w$				
xw				
xw'				



a)



e)



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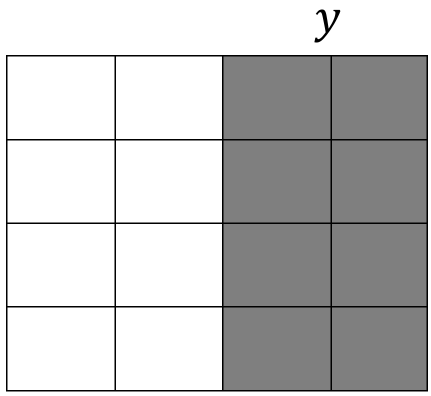

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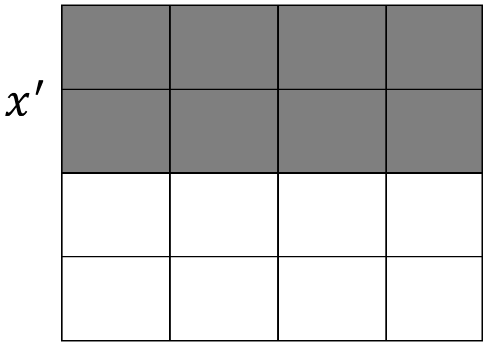
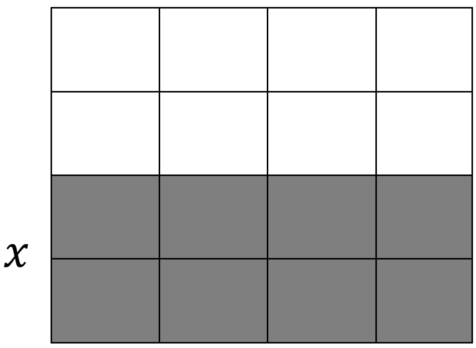
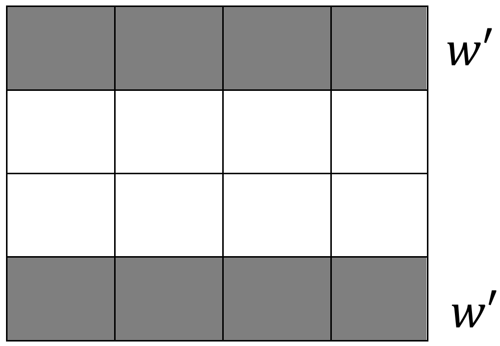


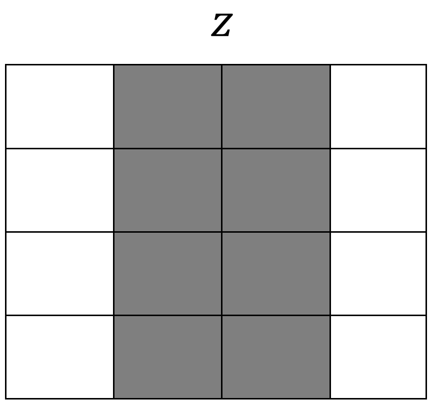
d)



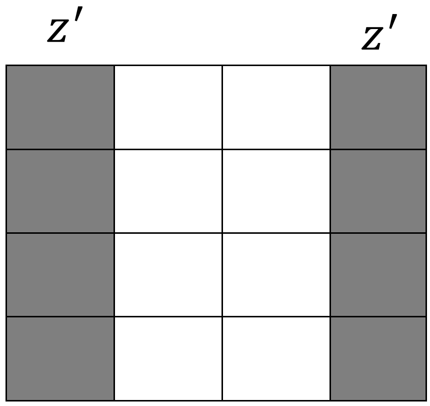
f)



g)



h)



Eg 18:

	$y'z'$	$y'z$	yz	yz'
$x'w'$	1			1
$x'w$		1	1	
xw		1	1	
xw'	1			1

Eg 19:

	$y'z'$	$y'z$	yz	yz'
$x'w'$	1	1	1	1
$x'w$				
xw			1	
xw'	1	1		

Eg 20: Construct karnaugh maps for the function for the following truth table. Use karnaugh map method to find a Boolean expression for the function f.

x	*y*	*z*	*w*	*f(x,y,z,w)*
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

	$y'z'$	$y'z$	yz	yz'
$x'w'$				
$x'w$				
xw				
xw'				